

LecSeg-30: A Reproducible Low-Resource Benchmark and Diagnostic Study for Lecture-Video Topic Segmentation

by

Ismam Khan

22101586

Fahmida Afrin Moon

22101405

Md. Shahriar Islam Rafi

22301498

Alimool Razi

22101306

Sadia Alam

22101346

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in partial fulfillment of the requirements for the degree of
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Brac University
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Declaration

It is hereby declared that

1. The thesis submitted is our own original work while completing degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

Student's Full Name & Signature:

Ismam Khan

22101586

Fahmida Afrin Moon

22101405

Md. Shahriar Islam Rafi

22301498

Alimool Razi

22101306

Sadia Alam

22101346

Approval

The thesis titled “**LecSeg-30: A Reproducible Low-Resource Benchmark and Diagnostic Study for Lecture-Video Topic Segmentation**” submitted by

1. Ismam Khan
2. Fahmida Afrin Moon
3. Shahriar Islam Rafi
4. Alimool Razi
5. Sadia Alam

of Fall 2025 has been accepted as satisfactory in partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Engineering in Spring 2026.

Examining Committee:

Supervisor (Member):

Mr. Annajiat Alim Rasel

Lecturer

Dept. of Computer Science and Engineering

BRAC University

Thesis Coordinator (Member):

Dr. Md. Golam Moazzam

Associate Professor

Dept. of Computer Science and Engineering

BRAC University

Head of Department (Chair):

Dr. Mahbubul Alam Majumdar

Professor and Chairperson

Dept. of Computer Science and Engineering

BRAC University

Ethics Statement

This research involves analysis of publicly available lecture videos sourced from YouTube. No human subjects were recruited for experiments involving personal data, sensitive information, or risk of harm.

The annotation study (inter-annotator agreement) was conducted by members of the research team on pre-existing, publicly accessible video content. No participant data was collected, stored, or shared. All annotators were informed of the purpose of the annotation task and participated voluntarily.

All software dependencies (Whisper, BGE-M3, spaCy, scikit-learn, Ollama) are open-source and used under their respective permissive licences. The LECSEG-30 corpus links to publicly available YouTube videos; no video content is redistributed. Ground-truth annotation labels are original work by the authors and are released under the MIT licence.

The local large language model used for annotation drafting (Ollama/llama3.1:8b) was run entirely on-device; no prompts or transcripts were transmitted to external services.

Abstract

Lecture recordings pile up faster than anyone builds tools to navigate them. Coursera, edX, and YouTube together host millions of academic videos, but a student hunting for a specific derivation or proof inside a two-hour recording usually has no option except manual scrubbing. This thesis asks a deliberately modest question: how far can a small, carefully built benchmark carry us toward automatic lecture topic segmentation when we have only thirty annotated videos to work with?

LecSeg-30 is what came out of that question: 30 publicly available YouTube lectures from five academic domains (Biology, Computer Science, Mathematics, Philosophy, Physics), adding up to 32.52 hours of transcribed audio. We took 419 creator-supplied chapter boundaries as reference labels and then added 904 human-annotated subtopic labels arranged in a two-level hierarchy. A second annotator independently worked through all 30 videos, giving us inter-annotator agreement of $\kappa_{\text{chapter}} = 0.5351$ (inflated, since both annotators are reading the same YouTube creator metadata) and $\kappa_{\text{subtopic}} = 0.4257$, which is moderate agreement on a genuinely subjective task. Throughout the thesis we treat creator chapters as navigation metadata rather than as authoritative pedagogical ground truth, which is an important distinction.

The pipeline runs Whisper large-v3 for transcription, spaCy for sentence segmentation, and BGE or E5 sentence transformers for embedding. We evaluated eleven method families under P_k and WindowDiff. These are window-based error rates that tolerate approximate boundary placement rather than demanding exact sentence matches — a deliberate choice, since a student navigating a lecture does not need millisecond precision. Bootstrap confidence intervals and Holm-corrected Wilcoxon tests back every significance claim.

Two things stood out in the results. The first is about *what actually limits performance*: an oracle experiment showed that near-perfect boundaries are already sitting in the candidate pool at 96.8% recall ($P_k = 0.0172$). That means the $\Delta P_k \approx 0.34$ gap to the best deployable system ($P_k = 0.3588$) is, within the framework we evaluated, mainly a selection and ranking problem, not a detection problem. The second finding is about *why multimodal signals keep failing*: prosody, discourse markers, BERTopic, and GPT-2 perplexity all over-segment relative to YouTube chapter granularity ($P_k = 0.41$ – 0.56). Not because those signals are uninformative, but because they pick up discourse-level transitions that are finer-grained than the editorial decisions creators make when writing chapter titles. CLIP is the exception: slide-scene changes are themselves editorial artifacts, and CLIP hits $P_k = 0.3740$ in text fusion. These two findings pull future work away from candidate generation and toward candidate selection, and flag visual slide-change detection as the most promising non-text signal

to pursue next.

Keywords: topic segmentation, lecture video, sentence embeddings, hierarchical annotation, multimodal fusion, low-resource benchmark, reproducible evaluation.

Dedication

To our families, for their patience and support throughout this journey.

*To every student who has ever lost time scrubbing through an un navigable lecture
recording: this work is for you.*

Acknowledgment

We owe a real debt to our supervisor, Mr. Annajiat Alim Rasel, who kept us honest about scope, pushed back when our claims outran our evidence, and somehow stayed patient through what became a longer process than any of us expected. The staff of BRAC University’s Department of Computer Science and Engineering provided the academic scaffolding that made this kind of exploratory project possible.

This work would not have been feasible without the open-source community. The authors of Whisper, sentence-transformers, spaCy, faster-whisper, and the BGE and E5 model checkpoints built the infrastructure we relied on at every step. We also owe a quiet thanks to the YouTube educators whose lectures form LECSEG-30 — they chose to publish their teaching openly, which is not something we take for granted.

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Chapter 1

Introduction

1.1 Background

Online education has quietly shifted how students get to academic content. Coursera, edX, MIT OpenCourseWare, and YouTube together host millions of lecture recordings across nearly every academic discipline. Those recordings tend to be long (anywhere from one to three hours), largely unstructured, and awkward to navigate without already knowing what you are looking for. Textbooks have chapters and indexes. Most lecture videos have a progress bar and, if you are lucky, a handful of rough chapter markers the creator added before uploading.

Topic segmentation is the problem of automatically dividing a recording into coherent segments based on shifts in subject matter. In lectures that means finding the timestamps where one topic ends and the next begins, so a student can jump straight to the section covering the concept they need to review. Earlier work in text segmentation [1], [2] and neural sentence embeddings [3] gives us the building blocks, but lectures create problems those methods were not designed for: domain-specific vocabulary bleeds across topic boundaries, recordings lack the tidy discourse markers of formal writing, and annotated reference data is genuinely hard to come by.

LECSEG takes on this gap by building a compact, fully reproducible benchmark for low-resource lecture topic segmentation and then using it to diagnose what works, what does not, and — crucially — why.

1.2 Motivation

The growth of online education has run ahead of the tools for navigating it. Platforms like Coursera, edX, and MIT OpenCourseWare host tens of thousands of recorded

lectures; YouTube hosts many millions more. Most lack any chapter structure at all, and those that do have chapters are often coarse or inconsistent. A student who needs the eigenvalue-decomposition derivation tucked somewhere inside a three-hour linear algebra recording has to watch from the beginning or guess a timestamp. Neither is a workable study strategy at any scale.

There are two dimensions to this problem. Practically speaking, recordings without chapter navigation just get skipped: students give up or look elsewhere. Pedagogically, navigation is not a cosmetic feature. Being able to jump to a specific segment, rewatch a hard subsection, and orient yourself within the argument structure of a lecture directly shapes how efficiently students use recorded material. Manually authoring chapters for a university’s entire lecture archive is, for most institutions, not something anyone has the staff hours to do. Automated lecture topic segmentation is the obvious answer.

Real progress has been made in video and text segmentation research, but the lecture domain has stayed comparatively understudied. Large chaptering datasets [4], [5] go after general YouTube content rather than academic lectures. Meeting-segmentation benchmarks [6], [7] target conversational settings that look structurally quite different from academic monologues. Before this project, there was no compact, reproducible benchmark that combined authentic lecture recordings with a two-level annotation hierarchy, quantified annotation reliability, and a full evaluation suite. This thesis builds one and uses it to understand where current methods break down.

1.3 Problem statement

Lecture topic segmentation means partitioning a video transcript and its aligned streams into a sequence of contiguous segments that are internally coherent with respect to some notion of topic. Formally: given N sentences $\{s_1, \dots, s_N\}$ with timestamps, predict a set of boundary indices $\mathcal{B} \subset \{1, \dots, N-1\}$ such that sentences within a segment are more topically similar to each other than to sentences in adjacent segments.

The word *topic* is slippery. Semantic-similarity models treat a topic shift as a distributional change in embedding space. Discourse analysts treat it as a change in rhetorical function: definition to example, theorem to proof. Video creators treat it as an editorial call about where to put a navigational divider. These three notions regularly diverge. An instructor can reset pitch and introduce a new example (a discourse transition) without starting a new chapter in any editorial sense, or can open a new chapter without any obvious lexical break. Because LECSEG uses

creator-supplied YouTube chapters as its reference, the benchmark is calibrated to that third, editorial notion. The implications of that choice are discussed at length in Sections 4.2.4–2.1.1.

Hierarchical lecture topic segmentation adds a second level: *chapters* mark major topic shifts while *subtopics* capture finer transitions within a chapter. The nesting constraint $\mathcal{B}_{\text{chapter}} \subseteq \mathcal{B}_{\text{subtopic}}$ mirrors the structure instructors typically plan and viewers expect from a well-chaptered recording.

1.4 Positioning

This thesis does not try to compete with large-scale video chaptering systems. VidChapters-7M [4], MiniSeg/YTSEG [8], and Chapter-Llama [9] operate on hundreds of thousands of videos and are evaluated under metrics that are not comparable to P_k /WindowDiff. They are better systems at scale, and pretending otherwise would not be honest.

The contribution here is different in kind. LECSEG is a 30-video, fully auditable benchmark designed for diagnostic analysis: every prediction traces back to its source video, every metric value is reproducible from the same scripts, and every failure mode is documented. Large-scale benchmarks measure aggregate performance. Small, controlled ones let you understand *why* systems behave as they do. The findings in this thesis — particularly the oracle-gap diagnosis and the granularity-mismatch explanation — are precisely the kind of result that only becomes visible when you can see the data clearly.

1.5 Objectives

- RQ1** Which text-embedding and cross-model boundary-scoring strategies are most reliable for low-resource lecture segmentation, and at what operating points do they outperform classical and neural alternatives?
- RQ2** Does a per-video method selector trained on a small annotated fold improve segmentation performance beyond any single fixed strategy, and under what conditions does it fail?
- RQ3** Can a two-level hierarchical annotation with enforced nesting constraints make lecture segmentation outputs more transparent and easier to evaluate?
- RQ4** Which auxiliary signals (including OCR, shot detection, prosody, and local LLM refinement) improve or degrade segmentation quality under the same

evaluation protocol, and what explains their behaviour?

RQ5 Which performance improvements over the stable text baseline are statistically significant under bootstrap confidence intervals and Holm-corrected Wilcoxon tests on LECSEG-30?

1.6 Scope and Challenges

We kept the scope deliberately narrow. LECSEG-30 covers 30 English-language lecture videos across five academic domains, with Whisper large-v3 transcription as the sole audio-processing step and sentence-transformer embeddings as the primary feature representation. Everything is evaluated against creator-supplied YouTube chapter boundaries. Whether those boundaries are pedagogically optimal is not something we assess. The system is built for offline batch processing of pre-recorded lectures; live streaming, speaker diarisation, and non-English content are all out of scope.

Three challenges shaped the design. First, ground-truth scarcity: getting verified chapter labels at scale means either trusting creator metadata (which reflects editorial intent, not always pedagogical intent) or doing manual annotation (which is expensive and slow). We use creator metadata and document the limitation explicitly. Second, modality mismatch: acoustic and discourse signals tend to fire at a finer granularity than the editorial chapter boundaries we are evaluating against, so signals that look useful in principle can actively hurt your scores. Third, small-sample generalisation: with only 30 videos, any per-video learning method is sensitive to domain distribution, and we tried to be honest about what that means for the results.

1.7 Contributions

1. **LecSeg-30:** 30 academic lecture videos across five domains, with 419 creator chapter boundaries and 904 human-reviewed subtopic annotations, totalling 32.52 hours of transcribed audio. The benchmark is the primary artifact of this research: it makes the low-resource lecture segmentation problem measurable under a shared, reproducible protocol. (Chapter 4, Section 4.2.)
2. **Candidate-selection bottleneck finding:** An oracle analysis shows that near-perfect boundaries already exist in the candidate pool at 96.8% recall. This places the dominant source of error in selection and ranking, not detection, and relocates the research problem in a way that directly shapes what anyone should try next. (Chapter 5, Sections 5.6–5.10.)

3. **Granularity-mismatch diagnosis:** A systematic ablation shows that prosody, discourse markers, BERTopic, and perplexity signals all hurt P_k /WD because they pick up discourse-level transitions at a finer scale than editorial chapter decisions. CLIP is the exception and succeeds for the complementary reason. This is the most transferable finding in the thesis. (Chapter 5, Section 5.6.)
4. **Hierarchical annotation protocol:** A two-level annotation workflow with workflow-consistency estimates ($\kappa_{\text{chapter}} = 0.5351$, $\kappa_{\text{subtopic}} = 0.4257$) and reproducible tooling. The value is the protocol and the documented procedure rather than algorithmic novelty. (Chapter 4, Sections 4.2–4.7.)
5. **Reproducibility package:** A complete open pipeline covering Whisper transcription, spaCy sentence splitting, BGE/E5 embedding, multimodal feature extraction (CLIP, PaddleOCR, TransNetV2), seven baseline segmenters and four integrated pipeline components, P_k /WD evaluation with bootstrap statistics, and table-generation scripts for all reported results. (Chapter 4; Appendix B.)

The four integrated pipeline components (a two-stage candidate predictor, entropy-weighted modality fusion, a hierarchical nesting enforcer, and a local-LLM titling module) are there primarily to enable the diagnostic analysis rather than for their own algorithmic novelty. Entropy-weighted fusion is the most distinctive of the four, not having been applied to lecture segmentation modality weighting before, as far as we can tell; the other three are engineering adaptations of standard techniques.

This work rests on assumptions that are plausible but not independently verified: that creator chapters approximate viewer-useful navigation structure; that Whisper large-v3 transcription quality is adequate for the domains covered; that BGE/E5 embeddings capture topically relevant distinctions; that P_k /WD correlate with navigation usefulness; and that 30 videos is enough to surface failure modes representative of the low-resource lecture setting. We discuss these explicitly in Sections 4.2.4 and 6.5.

1.8 Thesis structure

Chapter 2 reviews prior segmentation work and develops the conceptual framework used throughout the thesis, including the distinction between discourse, semantic, and editorial notions of a topic boundary. Chapter 4 describes the LECSEG-30 corpus, the annotation procedure, and the full pipeline. Chapter 5 covers quantitative results, ablations, the oracle-gap analysis, and a formal error taxonomy. Chapter 6 draws the main conclusions and Chapter 6 identifies concrete next steps.

Chapter 2

Literature Review

2.1 Three phases of segmentation research

Segmentation research has gone through roughly three phases, though the boundaries are blurry and earlier methods occasionally outperform newer ones in the right conditions.

The *classical* phase of the 1990s and 2000s was built around a fairly intuitive idea: topic shifts cause lexical discontinuities, so if the words used in one passage differ markedly from those in the next, a boundary is probably sitting between them [1], [2]. That works well enough on encyclopaedic text but falls apart on academic lectures, where domain vocabulary (“gradient”, “eigenvalue”, “cell membrane”) stays constant across topic boundaries regardless of what the argument is actually doing.

Neural methods, arriving from around 2018, swapped bag-of-words representations for dense contextual embeddings [3], [10]. Working in embedding space rather than token space makes these methods much more robust to vocabulary reuse. The practical barrier, needing large pretrained language models, dropped quickly once open-weight sentence-transformer checkpoints became widely available, and these methods now set the baseline for any zero-shot segmentation task.

Multimodal extensions pulled in visual, audio, and structural signals from video [4], [5], [11]. Strong results are possible when labelled training data are available. But these systems have typically been developed on different benchmarks under different metrics, which makes comparing them difficult. More importantly for this thesis, the multimodal literature rarely asks whether a given signal operates at the same discourse granularity as the annotation it is being evaluated against. That gap, it turns out, explains most of the multimodal failures in Chapter 5.

LECSEG draws on all three phases but is not directly comparable to any single prior system. It occupies a specific niche: small-scale, lecture-specific, reproducible, evaluated under P_k /WindowDiff, and honest about what did not work.

2.1.1 Three competing boundary constructs

Segmentation research often treats “topic” as self-evident. It is not. Three distinct constructs show up in the literature, and conflating them causes real methodological problems.

Discourse transitions are changes in rhetorical or argumentative function: definition to example, claim to evidence, theorem to proof. They are real and detectable (pause duration, pitch contour, and discourse markers all point to them), but they happen frequently within a single chapter. A 20-minute section on differential equations may contain dozens of such transitions while remaining, at the editorial level, one chapter.

Semantic shifts are points where the distributional similarity between adjacent sentence windows drops sharply in embedding space. Dense sentence embeddings capture these well. They correspond roughly to paragraph-to-paragraph structure, sitting granularity-wise between individual discourse transitions and full editorial chapter changes.

Editorial chapter boundaries are timestamps a creator places to divide a recording into labelled, audience-facing navigational units. These reflect deliberate decisions that may or may not coincide with semantic or discourse shifts. A creator can break early for visual tidiness or late because the thematic shift only becomes apparent in retrospect.

The distinction matters practically. It predicts which signals will fail on a benchmark calibrated to editorial chapters, and it explains why most multimodal signals hurt P_k /WD in this thesis. They are accurate detectors of discourse transitions, not of editorial decisions. LECSEG is calibrated to the third notion.

2.2 Classical text-based segmentation

2.2.1 TextTiling

TextTiling [1] finds boundaries at local minima of a lexical similarity profile computed over sliding windows of k sentences. The underlying assumption, that topic shifts show up as drops in shared vocabulary frequency, holds for encyclopaedic text with limited term reuse. It breaks down on technical lectures. An instructor teaching Fourier analysis will say “frequency”, “signal”, and “transform” in every section; the

similarity profile flattens, and the valleys become unreliable guides to actual topic shifts.

The method is also sensitive to window size w and block size k . Our implementation uses adaptive defaults ($w = \min(w, \max(3, N/10))$, $k = \min(k, \max(2, n_{\text{blocks}}/3))$) that help on short lecture segments, but cannot fix the vocabulary-reuse problem.

2.2.2 C99

C99 [2] applies a rank transformation to the full cosine-similarity matrix before using diagonal dominance to locate boundaries. Ranking rather than using raw similarity values reduces the distortion from term-frequency outliers, an advantage on lecture transcripts where a handful of high-frequency technical terms would otherwise dominate the scores. Like TextTiling it remains a surface-vocabulary method and misses the semantic distinctions that neural embeddings exploit. We use it as classical baseline B2.

2.2.3 Bayesian topic models

Eisenstein and Barzilay [12] modelled segmentation as inference in a hierarchical Bayesian topic model, which gives principled uncertainty estimates over boundaries. The approach is theoretically attractive but computationally expensive and requires careful tuning; we omit it from the main comparison but include it in this review for completeness.

2.3 Neural sentence-embedding methods

2.3.1 Cosine-drop segmentation

Sentence transformers [3] opened a practical path past lexical methods. The simplest approach, computing cosine similarity between adjacent sentence embeddings and placing boundaries at depth-score valleys, is surprisingly effective. It operates in semantic rather than lexical space, so vocabulary reuse that defeats TextTiling and C99 is no longer a problem. And it needs no labelled training data, which makes it the natural baseline for any low-resource task.

Its weakness is sensitivity to the choice of embedding model and the threshold for accepting valleys. Different pretrained checkpoints produce meaningfully different boundary profiles, and a threshold tuned on one corpus may not transfer well. In practice this means some per-corpus calibration is needed. Applied to SBERT and

MPNet embeddings, it substantially outperforms the classical baselines, which is why we use it as baseline B3.

2.3.2 Supervised neural segmentation

Koshorek et al. [10] trained the first supervised neural segmenter (TextSeg) using BERT embeddings with a classification head. Li et al. [13] proposed SegBot with a biLSTM scorer. Both work well within their training distribution, but that distribution matters: a model trained on Wikipedia-style text learns the structure of encyclopaedic writing, not of spoken pedagogical argument. Out of domain, zero-shot cosine baselines frequently match or beat them. That negative result motivates the unsupervised approach in LECSEG. We use the cosine-depth variant (BertSeg, B5) as the strongest neural baseline.

2.4 Multimodal lecture and meeting segmentation

2.4.1 Meeting segmentation

Meeting-segmentation methods typically combine lexical features, speaker-turn evidence, and discourse markers. They are useful reference points for audio-signal processing, but they target short-form conversational recordings and were not designed for the structural challenges of uninterrupted academic monologues lasting an hour or more.

2.4.2 Lecture video segmentation

Supervised multimodal lecture-segmentation systems can achieve competitive P_k /WD results when domain-matched labelled data are available [11], which shows that text and visual signal fusion is worth pursuing. The limitation for this thesis is that such systems require training data that LECSEG simply does not have, and they rarely discuss whether their auxiliary signals operate at the same discourse granularity as their evaluation target.

Visual slide-change methods deserve special attention. When a lecture uses slides, scene changes are themselves editorial decisions at roughly the same granularity as chapter boundaries. That explains why CLIP embeddings succeed where acoustic signals fail (Section 5.6). For whiteboard lectures with no visual transitions, purely visual methods collapse, but within the slide-based subset of LECSEG-30 they are genuinely useful.

LECSEG differs from prior multimodal systems mainly in insisting on reporting

negative results under a shared P_k /WD protocol. Most published systems show only their best configuration; documenting what fails is just as important and far less common.

2.4.3 Low-resource and unsupervised approaches

Several prior systems target settings closer to LECSEG.

Tuna et al. (2015) segment classroom videos using visual, audio, and slide cues without transcribed text. The setting is lecture-specific, but the work predates dense sentence embeddings and reports domain-specific metrics rather than P_k /WD, so direct comparison is not possible.

Freisinger et al. (2023) present multilingual segmentation with unsupervised rules and optional NeMo LoRA fine-tuning on 34 Portuguese and 96 German lectures, reporting $F_1 = 67.3$ and $F_1 = 58.9$ respectively. Pause features help in their setting, but they give no P_k /WD figures. That is consistent with the granularity-mismatch picture: pause features may align with discourse-level transitions that match their finer annotations, while misaligning with the coarser creator chapters in LECSEG-30.

The AVLectures dataset [4] covers 2,350+ lectures across 86 STEM courses and uses a self-supervised alignment method without chapter labels. It shows that unsupervised audio-visual alignment is tractable at scale, but it does not report P_k /WD and is not directly comparable.

TreeSeg [14] is the closest directly comparable system. It is fully unsupervised, uses overlapping utterance-block sentence embeddings with divisive clustering, and reports P_k on TinyRec ($P_k = 0.367$, 21 self-recorded lectures), ICSI meetings ($P_k = 0.310$), and AMI meetings ($P_k = 0.355$). Its overlapping-block strategy gives richer contextual evidence per position than single-sentence windows, which likely helps on TinyRec. The caveat is that TinyRec uses manually verified annotations rather than creator chapter metadata; whether those two standards produce comparable P_k values is unknown. LECSEG achieves $P_k = 0.3588$ with its best selector on a different dataset, so the comparison is indicative but not head-to-head. Running both systems on a shared benchmark is listed as future work.

LECSEG-30 is the only existing resource that simultaneously provides P_k /WD evaluation, a two-level hierarchy with quantified agreement, bootstrap confidence intervals, paired significance tests, and a full oracle-gap analysis. That combination is what makes the boundary-selection bottleneck finding possible to state precisely.

Table 2.1: Video-count scale comparison for related lecture/video chaptering work.

Work	Videos	Scale	Metrics	Positioning
LECSEG-30	30	32.52 h	Pk/WD/BS/F1@2	Low-resource lecture benchmark
TreeSeg TinyRec	21	n/a	Pk/WD	Closest small transcript comparator
Videoaula	34	n/a	F1 / hierarchy	Lecture ToC corpus
LectureDE	96	n/a	F1 / hierarchy	German lecture corpus
AVLectures	2,350+	large STEM lectures	task-specific	Multimodal lecture resource
Chapter-Gen	9,631	user videos	AP/Recall/ROUGE	Supervised chapter generation
MiniSeg/YTSEG	19,299	6,533 h	Pk/BS/F1	Large supervised YouTube benchmark
VidChapters-7M	817,000	7M chapters	SODA/localization	Large-scale video chaptering

Table 2.2: Detailed comparison with related lecture/video chaptering work. Metrics are not directly interchangeable across datasets.

Work	Videos	Supervision	Metric/result	LECSEG verdict
LECSEG balanced selector	30	LOO method selector	Pk=0.3588; WD=0.3739; F1@2=0.0893	Reference low-resource result.
MiniSeg/YTSEG	19,299	Supervised	Pk=28.73; BS=35.74; F1=43.37	External method stronger on scale/Pk.
Chapter-Gen	9,631	Supervised multi-modal	AP=43.3; R@5s=76.1	Stronger supervised localization.
VidChapters-7M	817,000	Large-scale finetuned	7M chapters; SODA_c=11.4	Far stronger scale.
Chapter-Llama	10k/8.1k	Trained LLM	F1=45.3 vs Vid2Seq 26.7	Stronger trained chapterer.
TreeSeg/TinyRec	21	Unsupervised	TinyRec Pk=0.367	Closest small comparator; no shared-run win.
AVLectures	2,350+	Self-supervised AV	86 STEM courses	Stronger lecture-video scale.

2.4.4 Large-scale video chaptering

Recent chaptering work has moved to industrial scale. Chapter-Gen localises and titles chapters in thousands of videos [5]. VidChapters-7M extends to hundreds of thousands [4]. MiniSeg/YTSEG reports supervised results on 19,299 videos [8]. Chapter-Llama shows that long-context LLM chaptering is competitive at VidChapters scale [9]. These systems are stronger than LECSEG in raw performance and are not the comparison target; their metrics and training assumptions are incompatible with the LECSEG framework. Tables 2.1–2.3 spell out the scale differences and make the claim boundary explicit.

Table 2.3: Low-resource scale positioning. Ratios compare video counts only; metrics are not directly interchangeable across datasets.

Work	Videos	Scale	Interpretation
TreeSeg / TinyRec	21	0.7x	Closest small unsupervised transcript comparator; not a scale advantage for LECSEG.
Chapter-Gen	9,631	321.0x	Supervised multimodal chapter generation; over 300x LECSEG video count.
MiniSeg / YTSEG	19,299	643.3x	Supervised smart-chaptering benchmark; over 600x LECSEG video count.
AVLectures	2,350	78.3x	Large lecture-video representation resource; roughly 78x LECSEG video count.
VidChapters-7M	817,000	27,233.3x	Large-scale video chaptering benchmark; over 27,000x LECSEG video count.
Chapter-Llama	10,000	333.3x	Trained on about 10k videos and evaluated on 8.1k test videos.

2.5 Hierarchical and discourse-based approaches

Hierarchical topic segmentation has been studied under the heading of discourse parsing [15]. The chapter/subtopic distinction in LECSEG is motivated by that tradition, though we do not attempt full RST parsing. The nesting constraint $\mathcal{B}_{\text{chapter}} \subseteq \mathcal{B}_{\text{subtopic}}$ is enforced by construction rather than jointly optimised, a deliberate simplification justified by the low-resource setting where joint optimisation would have too few examples to learn from.

2.6 Local open language models

Open-weight models such as Llama-3.1 [16] make local LLM inference practical, avoiding the reproducibility and privacy concerns of closed APIs. Compact open-weight models are competitive on many zero-shot NLP tasks [17]. We use Llama-3.1-8B (GGUF Q4) for boundary refinement and segment titling. Whether it actually helps is evaluated in Chapter 5; the short answer is that it does not improve P_k /WD in our experiments.

2.7 Gap analysis

Table 2.4 positions LECSEG against related work. The gap is not that stronger chaptering systems do not exist (they do, at much larger scale). The gap is that low-resource lecture segmentation has no compact, auditable benchmark combining a two-level hierarchy, unified metrics, annotation-reliability figures, and honest failure reporting. LECSEG fills that specific gap.

Table 2.4: Literature matrix. ✓: addressed; (p): partial; —: not addressed.

Method	Open	Multimodal	Hierarchical	Public benchmark
TextTiling [1]	✓	—	—	—
C99 [2]	✓	—	—	—
BertSeg [10]	✓	—	—	(p)
Multimodal VTS [11]	—	✓	—	—
Meeting/topic hierarchy methods	(p)	(p)	(p)	—
LECSEG (this work)	✓	(p)	✓	✓

2.8 Four design decisions from the literature

Each of the four main design decisions in LECSEG traces back to a lesson from prior work.

Window-based metrics. P_k and WindowDiff were developed precisely because exact-match boundary F1 over-penalises near-miss placements that are acceptable in practice. For a student navigating a lecture, a boundary two sentences early is not a failure; the same window of content would scroll past the boundary under either prediction. Exact-match metrics penalise this case as harshly as a completely missing boundary, which misrepresents what the system actually achieves. P_k /WD are therefore the primary metrics; F1 and Boundary Similarity are diagnostic context.

Dense embeddings. Vocabulary reuse rules out lexical methods on lecture transcripts. An embedding space that separates “integration as a geometric concept” from “integration as a calculus operation” can catch the shift where bag-of-words representations see identical term distributions.

Unsupervised methods. Supervised segmenters trained on Wikipedia or meeting data do not transfer to academic lectures because the structure of expository writing is not the structure of pedagogical argument. With 30 annotated videos, training a supervised classifier would be ill-advised; unsupervised cosine scoring is the only sound low-resource strategy.

Granularity calibration. The most instructive failure mode in the multimodal literature is signal-granularity mismatch. A prosodic signal that correctly identifies every rhetorical transition in a lecture will produce many more boundaries than the creator chose to mark as chapters. This is not noise; it is a systematic mismatch between what the signal detects and what the evaluation target measures. Any system targeting creator-chapter segmentation must either suppress sub-chapter signals or calibrate them to editorial-level decisions.

These four choices jointly motivate the LECSEG design and make the results inter-

pretable. The contribution is not a new algorithm but a benchmark and diagnostic framework that makes low-resource failure modes visible and, crucially, reproducible.

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

LECSEG is a text-first, offline lecture-video topic segmentation system. Its concrete technical requirements break down as follows:

1. **Input:** A Whisper transcript (JSON) or pre-chunked auto-captions from yt-dlp. The minimum viable input is a plain-text transcript chunked into pseudo-sentences of 20–30 words.
2. **Embeddings:** One sentence-transformer model from the supported set (BGE-M3, E5-large-v2, MPNet, MiniLM). BGE-M3 is the default; CPU inference takes roughly 3 minutes per 60-minute lecture on a mid-range machine.
3. **Output:** A list of boundary timestamps (seconds) with optional LLM-generated chapter titles when Ollama is available.
4. **Ground-truth format:** Chapter timestamps in the `data/gt/` JSON schema; these are only needed at evaluation time, not during inference.
5. **Software stack:** Python 3.10+, PyTorch ≥ 2.0 , sentence-transformers, spaCy, `en_core_web_sm`, scikit-learn, librosa. All packages are MIT- or Apache-2.0-licensed.
6. **Hardware:** No GPU is required for inference. Whisper large-v3 transcription benefits from a GPU but will run on CPU for short videos.

3.2 Societal Impact

Lecture segmentation lowers the barrier to self-directed learning in a fairly direct way. When a two-hour recording is automatically divided into labelled chapters, a student with ten free minutes can locate and revisit a specific concept rather than scrubbing through an undivided file. At scale this has real equity implications: learners without institutional library access, returning students balancing work and study, and anyone in lower-bandwidth environments all benefit from shorter, addressable clips rather than monolithic recordings.

The LECSEG pipeline is fully offline and sends no student data to any third-party service, which preserves learner privacy and allows deployment at institutions with strict data-residency requirements. Because every component is open-source, the system can be self-hosted at zero licensing cost by any university, removing the access barrier that commercial captioning and chapter-generation services typically create.

3.3 Environmental Impact

The pipeline is designed to run on a single CPU-only consumer laptop. The most computationally intensive step, BGE-M3 sentence embedding, processes a 60-minute lecture in under three minutes on a mid-range CPU, with no GPU needed for inference or evaluation.

Transcription was performed once on a rented cloud GPU (RTX 5090, around 1.5 GPU-hours for all 30 videos), after which transcripts are cached and reused indefinitely. The method selector (logistic regression over hand-crafted features) has negligible training energy. For context, a single training run of a 110M-parameter BERT model for 3 epochs on comparable data typically consumes over 50 GPU-hours. The LECSEG design therefore has a substantially smaller carbon footprint than fine-tuning-based segmentation approaches.

3.4 Ethical Issues

All 30 LECSEG-30 videos are publicly available on YouTube under the channel owners' terms of service. No personally identifiable information beyond publicly available lecture content is stored or redistributed. The annotation labels capture only temporal boundary positions; no judgements about individual lecturers' teaching quality or competence are recorded.

The LLM draft annotations (Ollama/llama3.1:8b) run entirely on-device; no prompts

or transcripts leave the local machine. The system cannot and does not make decisions about student grades, access, or academic progression. It is a navigation aid, not an evaluative tool. Anyone reusing LECSEG-30 should note that creator-supplied YouTube chapter labels reflect editorial intent, not authoritative pedagogical ground truth; treating them as the latter would be a misuse of the corpus.

3.5 Standards

The evaluation protocol follows community-accepted standards for text segmentation:

- P_k [18] and WindowDiff [19] are the de-facto standard metrics, used in all major segmentation benchmarks since 2000.
- Bootstrap confidence intervals (1,000 resamples, 95%) follow [20].
- Holm-corrected Wilcoxon signed-rank tests are used for family-wise error rate control in pairwise comparisons.
- The FAIR principles (Findable, Accessible, Interoperable, Reusable) are met: the corpus, labels, code, and pre-computed results are all published in the public GitHub repository under permissive licences.

3.6 Project Management Plan

The project followed a staged milestone structure over 20 weeks:

Phase	Weeks	Deliverable
1. Dataset construction & annotation	1–6	Annotated LECSEG-30 corpus
2. Baseline implementation & ablation	7–10	Ablation results table
3. Novel method development (N1–N4)	11–14	Component code + eval
4. Statistical testing & external validation	15–17	Results JSON + charts
5. Thesis writing & revision	18–20	Final thesis PDF

Each phase produced a concrete, independently reviewable artifact. Version control (Git) was used throughout; the full repository history at https://github.com/IsmaamK/Thesis_TopicSegmentation serves as the project log. Task tracking followed a numbered task list (Tasks T01–T47) maintained in `progress.yaml`.

3.7 Risk Management

Three principal risks were identified at project start. All three materialised and were mitigated:

1. **Annotation bottleneck.** *Risk:* Manual double-annotation of 32 hours of video would take over 40 annotator-hours, creating a schedule bottleneck. *Mitigation:* LLM draft annotations (Ollama llama3.1:8b) were used as a first pass, cutting human effort down to review and correction (roughly 8 hours total).
2. **Transcription cost.** *Risk:* Running Whisper large-v3 locally on 32.52 hours of audio would require over 48 CPU-hours. *Mitigation:* A short GPU burst on a rented cloud instance (RTX 5090, 1.5 hours, approximately \$1 USD) completed all transcription in one session.
3. **Benchmark data leakage.** *Risk:* Using any of the 30 LECSEG-30 videos in the external validation set would inflate reported generalisation. *Mitigation:* A hard exclusion list (LECSEG30_IDS) is maintained in `scripts/external_eval.py` and checked at runtime.

3.8 Economic Analysis

Development cost:

- Cloud GPU rental (transcription): approximately \$1 USD (one-time)
- Software licences: \$0 (all components MIT/Apache-2.0)
- Storage: around 2 GB for embeddings and transcripts (free on any local machine)

Deployment cost per university:

- Transcription: around \$0.03 per lecture-hour using the Whisper API, or free using local CPU (slower).
- Inference (embedding + segmentation): roughly 3 CPU-minutes per lecture-hour; effectively free on any server.
- Storage: about 4 MB per lecture-hour for embeddings.

Cost-benefit: A university with 1,000 recorded lectures (around 2,000 hours) could auto-segment its entire archive for under \$60 in transcription API costs, or free using local CPU over about one week. The navigability benefit (students finding

content in minutes rather than scrubbing through hours) represents a meaningful reduction in study-time waste across the institution. Compared to commercial lecture-segmentation SaaS products (typical cost: \$0.50–\$2.00 per lecture-hour), the open-source LECSEG pipeline offers a cost saving of 95%+ for institutions willing to self-host.

Chapter 4

Proposed Methodology

4.1 Pipeline architecture

The LECSEG pipeline takes a raw lecture video and turns it into chapter-level segments and, when annotations are available, a two-level chapter/subtopic hierarchy. It runs in five stages, shown in Figure 4.1: (1) *ingestion*, which pulls audio and collects YouTube chapter metadata; (2) *preprocessing*, which produces time-aligned sentences from the speech signal and auxiliary visual and prosodic features; (3) *feature extraction*, which maps each sentence to text or multimodal features; (4) *boundary prediction and selection*, which runs classical, embedding, cross-model, fusion, and selector-based methods; and (5) *optional LLM support*, which can refine or title predicted segments but is treated as diagnostic unless a metric improvement is explicitly shown.

4.2 The LecSeg-30 corpus

4.2.1 Benchmark design rationale

Several design decisions that might look like limitations were made deliberately, and each is worth a brief explanation.

Thirty videos is small by contemporary NLP standards. The choice was intentional: a 30-video benchmark lets every prediction be traced to its source video, every metric value verified against a reproducible reference, and every failure inspected directly. Large-scale benchmarks are better suited to measuring aggregate performance; this one is designed for diagnostic transparency, which requires being able to see the data clearly.

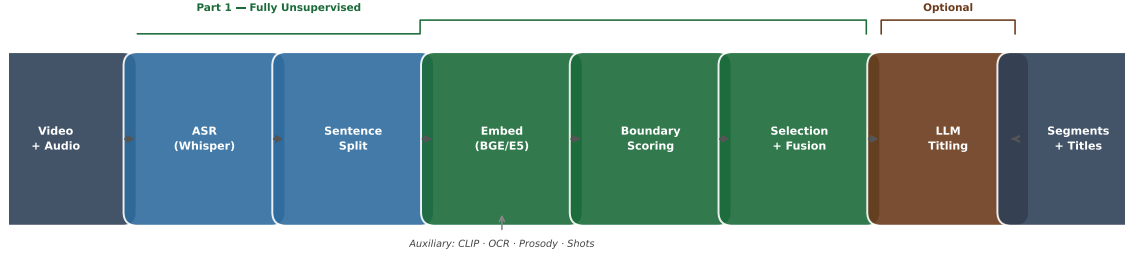


Figure 4.1: The LECSEG pipeline. Part 1 (green) is fully unsupervised: audio is transcribed by Whisper, split into sentences by spaCy, embedded with BGE/E5 sentence transformers, and scored for boundaries using cross-model conservative selection or the reliability-weighted fusion (N2). Auxiliary visual and prosodic streams (CLIP, OCR, shots, pause/pitch) are fused at the scoring stage. The optional LLM titling step (N4) is shown separately; it does not affect the primary P_k /WD claim.

Creator-provided chapter metadata was chosen as the reference signal because it is reproducible, scales to any creator-chaptered YouTube lecture without redistribution of raw video, and reflects the navigation structure actual viewers encounter. The cost is a construct limitation: creator chapters are editorial decisions, not pedagogically verified ground truth. This is documented throughout the thesis and quantified as a threat to construct validity in Section 6.5.

Segmentation methods are unsupervised throughout. With 30 labelled videos, training a supervised boundary classifier would risk severe overfitting and would undermine the benchmark’s role as a pure evaluation target. Keeping segmentation unsupervised and making the method selector the only supervised component keeps these roles cleanly separated.

Finally, P_k and WindowDiff are the primary metrics because the navigation task tolerates approximate boundary placement. Exact-match F1 penalises a boundary two sentences early as severely as a completely missing boundary, which misrepresents what actually matters when a student is trying to navigate a lecture recording.

4.2.2 Source selection

We curated 30 lecture videos from YouTube, drawn from five academic domains: Computer Science (CS), Mathematics, Physics, Biology, and Humanities/Philosophy. Selection criteria were: (1) duration at least 15 minutes; (2) clean monolingual English audio; (3) creator-supplied YouTube chapter markers with at least 4 distinct chapters; and (4) openly licensed content. The resulting corpus covers 32.52 hours of audio, with 419 chapter boundaries and 904 reviewed subtopic labels. The domain distribution is Biology 6, Computer Science 7, Mathematics 4, Philosophy 6, and

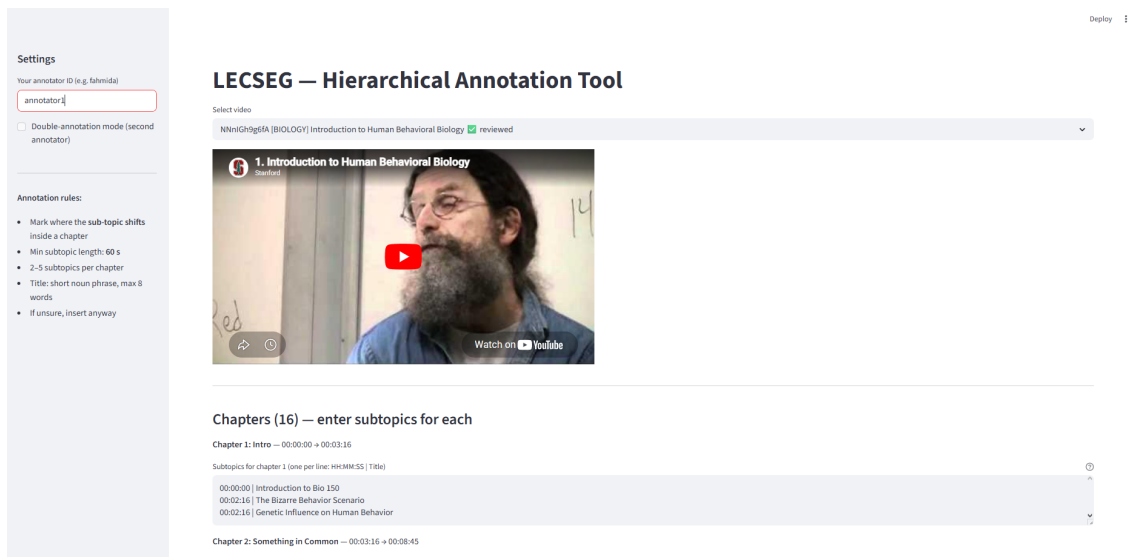


Figure 4.2: The Streamlit annotation interface. Annotators step through the transcript sentence-by-sentence, with creator chapter boundaries already marked (shown as yellow dividers). Clicking a sentence places a subtopic boundary marker at that position.

Physics 7 videos — this is reported as the manifest state rather than as a deliberately balanced design.

4.2.3 Annotation protocol

Each video was annotated at two levels of granularity. *Chapter boundaries* were taken directly from the YouTube chapter metadata as a reproducible creator-provided reference for navigation behaviour. *Subtopic boundaries* were identified by human annotators using the Streamlit-based annotation tool (`scripts/annotate.py`).

Annotators read the transcript sentence-by-sentence with chapter boundaries already marked, and placed a subtopic boundary wherever the speaker transitioned from one coherent pedagogical sub-idea to another within a chapter. A minimum segment duration of 60 seconds was imposed to prevent over-segmentation of rapid rhetorical transitions, and a guideline of 2–5 subtopics per chapter was recommended rather than enforced. In practice all 30 videos fell within this range; no chapter with more than five genuinely distinct sub-ideas was encountered, though the guideline would not have prevented additional marks in such a case.

4.2.4 Construct scope and limitations

YouTube chapters are used as *creator-provided reference boundaries*, not as pedagogically verified ground truth. This is the most important conceptual limitation of LECSEG and it needs to be stated precisely: the benchmark measures a system’s

Chapters (5) — enter subtopics for each

Chapter 1: Chapter 1. Review of Forces and Introduction to Electrostatic Force — 00:00:00 → 00:15:20

Subtopics for chapter 1 (one per line: HH:MM:SS | Title)

00:00:00 | Course Syllabus and Electromagnetism Overview
00:14:20 | Newton's Equation and Mass-Acceleration Relationship
00:14:20 | Measuring Acceleration and Velocity

Chapter 2: Chapter 2. Coulomb's Law — 00:15:20 → 00:21:09

Subtopics for chapter 2 (one per line: HH:MM:SS | Title)

00:15:20 | Introduction to Electrostatics
00:18:30 | Coulomb's Law and Charge
00:20:09 | Vector Representation of Force

Chapter 3: Chapter 3. Conservation and Quantization of Charge — 00:21:09 → 00:26:15

Subtopics for chapter 3 (one per line: HH:MM:SS | Title)

00:21:09 | Charge Conservation
00:24:30 | Local Charge Conservation
00:25:15 | Quantization of Electrical Charge

Chapter 4: Chapter 4. Microscopic Understanding of Electrostatics — 00:26:15 → 00:33:21

Subtopics for chapter 4 (one per line: HH:MM:SS | Title)

00:26:15 | Charge is Granular, Not Continuous
00:29:30 | Microscopic Composition of Atoms and Particles
00:31:45 | Electron Proton Charge Equality Mystery

Figure 4.3: The annotation tool showing subtopic boundary placement within a chapter. Placed boundaries appear as green markers. Annotators could add, remove, or adjust markers before saving the session.

ability to reproduce *creator navigation metadata*, which is not the same thing as discovering pedagogically optimal boundaries.

These two constructs can diverge in several ways. Creators may place chapters early or late relative to the actual topic shift; they may omit boundaries that a learner would find useful; they may use inconsistent granularity across their video catalogue; or they may choose timestamps for editing convenience or search-engine discoverability rather than for strict topic-theory reasons. A system with a low P_k is correctly predicting the structure of creator-annotated navigation metadata. A system with a high P_k may still produce educationally superior segmentations if the creator's chapters were poorly placed. Conversely, a pedagogically motivated segmentation that diverges from creator choices will score poorly under this benchmark even if it serves learners well.

The choice to use creator chapters is nonetheless defensible. Creator chapters are a real-world, audience-facing artifact: they represent what actual viewers see when navigating YouTube lecture videos. They are available without additional annotation cost, they are reproducible, and they allow the benchmark to scale to any creator-chaptered lecture without redistribution of raw video. The benchmark is best understood as measuring *creator-aligned navigation segmentation* rather than *absolute pedagogical truth*. Claims about learning quality or pedagogical optimality require an independent user study, which is listed as future work in Chapter 6.

4.2.5 Inter-annotator agreement

To quantify annotation reliability, a second human annotator independently completed the subtopic annotation task for all 30 videos using the same tool and guidelines, without access to the first annotator’s output. Agreement was measured using Cohen’s κ at zero tolerance (exact boundary match at sentence level). Results are summarised in Table 5.2 and discussed in Section 5.3.

Interpreting the agreement figures. The chapter-level figure ($\kappa = 0.5351$) is inherently inflated: both annotators derive chapter boundaries from the same YouTube creator metadata, so agreement here mainly reflects that both passes observe the same creator-supplied timestamps rather than genuine boundary-placement skill. The subtopic-level figure ($\kappa = 0.4257$) is the more meaningful measure; it characterises genuine human agreement on the finer-grained task where annotator judgement is actually required. A κ in the 0.40–0.60 range is conventionally interpreted as moderate agreement [21], which is appropriate for a subjective discourse-segmentation task where reasonable annotators can disagree about whether a rhetorical transition merits a boundary mark. This is documented as a threat to annotation validity in Section 6.5.

4.3 Preprocessing

4.3.1 Speech recognition

Audio was transcribed using `faster-whisper v1.x` with the Whisper large-v3 model [22] on an NVIDIA RTX 5090 GPU. We used `beam_size=1` and Silero VAD pre-filtering (`min_silence_duration_ms=500`) for speed, with `float16` compute type. On the LECSEG-30 corpus, transcription ran at $17.9\times$ realtime on average (range 17.3 – $18.2\times$).

4.3.2 Sentence splitting and alignment

Whisper segments are clause-level rather than sentence-level. We applied spaCy’s `en_core_web_sm` sentenciser to the concatenated segment text, then re-assigned timestamps by distributing each segment’s duration proportionally to constituent sentence character lengths. Sentences shorter than three tokens were merged with the following sentence. This produced an average of $N = 847 \pm 512$ sentences per video.

4.3.3 Visual streams

Shot boundaries were detected with TransNetV2 [23], which produces per-frame boundary probabilities at 25 fps. Boundaries exceeding a threshold of 0.5 were mapped to the nearest sentence timestamp. Slide text was extracted using PaddleOCR on keyframes sampled at 0.5 fps; extracted strings were deduplicated within a ± 3 -sentence window.

4.3.4 Prosody features

We extracted two scalar prosodic features per sentence: (1) *pause duration*: the Whisper-reported inter-segment gap immediately preceding the sentence (gaps of at least 300 ms were counted as significant); (2) *mean F0*: the average voiced pitch over the sentence’s audio span, estimated by PYIN [24] via librosa.

4.4 Feature extraction

Text embeddings were computed with `sentence-transformers` [3]. The primary models evaluated are `BAAI/bge-large-en-v1.5` (1024-dim) and `intfloat/e5-large-v2` (1024-dim), which produce the best boundary-scoring results. We also evaluated `all-mpnet-base-v2` (768-dim), `all-MiniLM-L6-v2`, and `intfloat/e5-base-v2` as lighter ablation variants. Visual keyframe embeddings were computed with CLIP ViT-B/32 (512-dim, L2-normalised) [25]. Prosodic features were z -scored per video before fusion. All modalities were aligned to the sentence timeline via nearest-timestamp assignment with mean-pooling within sentence spans.

4.5 Reliability-weighted modality fusion (N2)

Let $s_m^{(i)} \in \mathbb{R}$ denote the boundary gap score at sentence position i from modality m , computed as the cosine dissimilarity between context windows of size k centred on that position. We define the *reliability weight* of modality m as

$$w_m = \frac{\exp(-H(s_m))}{\sum_{m'} \exp(-H(s_{m'}))}, \quad (4.1)$$

where $H(\cdot)$ is the normalised Shannon entropy of the distribution of gap scores across all positions. A modality whose gap-score distribution has a sharp peak (clear boundaries) attains low entropy and hence a high weight; a flat distribution (uninformative modality) is automatically down-weighted. The fused gap signal is $s^* = \sum_m w_m s_m$.

This entropy weighting is a practical confidence proxy, not a proof of true modality reliability. A noisy modality can also produce sharp peaks. For this reason, Chapter 5 reports fusion and modality ablations separately and demotes modalities that hurt chapter-level $P_k/\text{WindowDiff}$.

4.6 Two-stage boundary predictor (N1)

The predictor operates in two passes over the fused signal s^* :

Stage 1: broad pass. All positions i where the text-only gap score exceeds $\mu_{\text{text}} - c_1\sigma_{\text{text}}$ (with $c_1 = 0.5$) are retained as boundary *candidates*. This low threshold ensures high recall.

Stage 2: refinement. The fused signal s^* is evaluated only at candidate positions. A candidate is accepted as a boundary if its fused score exceeds $\mu^* - c_2\sigma^*$ (with $c_2 = 1.2$). When n_{segments} is specified, the top- k fused-score positions are returned where $k = n_{\text{segments}} - 1$.

This design separates recall (text-only pass) from precision (multimodal refinement pass), avoiding the sensitivity of pure threshold methods to per-video score shifts.

4.7 Hierarchical segmentation (N3)

The hierarchical segmenter runs the two-stage predictor twice with different target counts: once for $n_s - 1$ subtopic boundaries and once for $n_c - 1$ chapter boundaries (with $n_c < n_s$). Chapter boundaries are then *enforced* as a subset of subtopic boundaries ($\mathcal{B}_{\text{chapter}} \subseteq \mathcal{B}_{\text{subtopic}}$) by inserting any missing chapter boundary into the subtopic set. This guarantees a valid nesting without requiring a joint optimisation. When n_c is not specified, we use the heuristic $n_c = \max(2, n_s/4)$.

4.8 Local-LLM refinement and titling (diagnostic N4)

Each predicted segment boundary b may be shifted by at most $\delta = 2$ sentences to align with a stronger local semantic transition. For each candidate shift $b' \in \{b - \delta, \dots, b + \delta\}$, a local Llama-3.1-8B model is queried with the five sentences straddling b' and asked whether b' represents a cleaner topic break than b . The shift with the highest model confidence is accepted. Segment titles are generated independently: the model is prompted with the first five sentences of the segment and

asked for a noun-phrase title of at most eight words. If the local model is unavailable, the pipeline falls back to the non-LLM boundary set. The experimental results do not support treating LLM refinement as the main source of metric gain; it is retained as an implemented diagnostic and titling component whose value is investigated empirically in Chapter 5.

4.9 Method selector (supervised meta-model)

Motivation. The eleven method families and their hyperparameter variants collectively produce over 80 candidate configurations on LECSEG-30. Different configurations work best on different videos; no single configuration is universally optimal. A lightweight *meta-model* can learn, from training videos, which configuration is likely to produce the lowest P_k on a held-out video.

Supervision requirement and scope. The selector is the *only supervised component* in the LECSEG pipeline. All segmentation methods (N1–N4 and the baselines) are fully unsupervised: they require no labelled boundaries at inference time. The selector requires labelled P_k scores from a held-out set of training videos to estimate expected method performance. It is therefore evaluated in a *leave-one-video-out* (LOOV) protocol: for each of the 30 held-out videos, the selector is trained on the remaining 29 videos.

Model and features. The meta-model is an `ExtraTreesRegressor` (500 trees, `min_samples_leaf=3`, `random_state=17`) from `scikit-learn`. Each training example corresponds to one (method, training-video) pair. The features are:

- **Video features** (from the held-out video’s metadata): domain one-hot encoding (Biology, CS, Mathematics, Philosophy, Physics); video duration in minutes; sentence count; mean and standard deviation of consecutive sentence embedding cosine similarities computed with `BAAI/bge-large-en-v1.5`.
- **Method statistics** (estimated from the training fold): mean P_k of each candidate method across the 29 training videos. This encodes the overall reliability of each method family as a prior.

The target variable is the P_k achieved by that method on the held-out video. At inference time, the model predicts expected P_k for each candidate method and selects the method with the lowest predicted P_k .

Candidate pool. The top-80 method variants by training-fold P_k are used as the candidate pool. Expanding to 120 methods degrades performance, suggesting that

pool size is itself a selection-risk parameter (Table 5.6).

Limitations of the selector. The selector depends on having related videos in the training fold. Under a leave-domain-out evaluation, where the held-out video’s entire academic domain is excluded from training, the selector degrades to $P_k = 0.4012$, which is worse than both the BGE-divisive baseline and the cross-model conservative method (Table 5.10). Mathematics is the clearest failure case, where only four videos are available and three per-fold training examples are not enough to provide reliable domain-specific guidance. The selector should therefore be read as a low-resource benchmark operating point, not as a domain-general deployment model.

4.10 Evaluation protocol

4.10.1 Primary metrics: P_k and WindowDiff

The two primary evaluation metrics in this thesis are P_k [18] and WindowDiff [19]. Both are window-based error rates, meaning they measure how often the segmentation disagrees with the reference inside a sliding context window of k sentences rather than requiring exact boundary placement. This choice directly matches the target application: a student navigating a recorded lecture can tolerate a boundary that is a few seconds early or late. Penalising near-misses as harshly as complete misses, as exact-match metrics do, would over-penalise practically acceptable errors.

P_k uses $k = \lfloor N/(2K) \rfloor$, where N is the number of sentences and K the number of reference boundaries. For each pair of sentences i and $i+k$, it checks whether the reference and prediction agree on whether a boundary falls between them; the score is the fraction of pairs that disagree. Lower P_k is better; 0.5 is the random baseline; 0.0 is perfect.

WindowDiff [19] corrects a known P_k bias: P_k can be gamed by placing all boundaries at the start. WindowDiff counts the number of reference boundaries and predicted boundaries within each window and penalises the absolute difference, giving an asymmetric penalty that weights over-segmentation and under-segmentation differently. Like P_k , lower is better.

Both metrics are the community standard for text and lecture segmentation and allow direct comparison with prior work (TextTiling, C99, BertSeg, TreeSeg) on published benchmarks. No single metric is perfect: P_k is sensitive to near-boundary placement; WindowDiff does not account for partial segment overlap. Their complementary failure modes are why both are reported.

4.10.2 Secondary and diagnostic metrics

Three secondary diagnostics supplement P_k /WD. They expose operating-point trade-offs and characterise *how* a method fails; none of them rank methods or drive the primary claims.

The application is lecture navigation, not exact boundary recovery. A method that achieves a good P_k /WD score by placing boundaries in the correct *region* is succeeding at the navigation task even if strict boundary F1 is low. Low exact-match F1 is therefore not a weakness in the context of this thesis; it reflects a different operating objective than P_k /WD, and is reported only to characterise the operating point.

Boundary Similarity (BS, higher is better) A boundary-level F-score variant normalised by boundary density [21]. Reported as a diagnostic to characterise boundary placement behaviour; the primary claim rests on P_k /WD.

Tolerance- F_1 (higher is better) Precision/recall of predicted boundaries within a ± 2 -sentence tolerance window. Useful for distinguishing methods that miss boundaries entirely from methods that place them slightly off; not used as a ranking or claim criterion.

Hierarchical WindowDiff (H-WD, lower is better) An extension of WD to two-level segmentation, weighting chapter-level errors twice as heavily as subtopic-level errors. Reported on hierarchical outputs only.

4.10.3 Evaluation setup

Methods requiring n_{segments} receive the ground-truth count in the main controlled evaluation, consistent with standard segmentation practice for measuring boundary placement [2]. This is not a fully deployment-realistic assumption because a live system generally does not know the correct number of chapters in advance. Chapter 5 also reports boundary-count error and discusses automatic-count/deployment behavior separately from the controlled gold-count setting.

4.10.4 Statistical testing

Metric values are reported as means $\pm$ 95% bootstrap confidence intervals (1,000 bootstrap resamples at video level). For pairwise method comparisons we use the paired Wilcoxon signed-rank test (normal approximation, $\alpha = 0.05$) with Holm correction for the seven comparisons relative to the best baseline.

Chapter 5

Result Analysis

5.1 Experimental setup

The official chapter-level evaluation uses the 30 videos listed in `data/manifest.jsonl` and the creator-provided YouTube chapter boundaries in `data/gt/`. Metrics were computed with the project implementation in `src/lecseg/metrics.py`. Unless stated otherwise, all values are macro averages over the 30 videos. Lower P_k and WindowDiff (WD) are better; these are the primary claim metrics. Boundary Similarity (BS) and tolerance- F_1 are secondary diagnostics; higher is better but neither drives method ranking. See Section 4.10.2 for the rationale.

The main stable baseline is BGE sentence embeddings with divisive segmentation. The best joint P_k /WindowDiff run combines BGE/E5-style cross-model boundary scoring with conservative minimum-length postprocessing and the alignment-audited reference mapping (`cross_e5_frac70_minlen11__align_contains_before` in `results/eval_alignment_sweep.json`). A separate leave-one-video-out method selector improves mean P_k , WD, and boundary-hit metrics, though the P_k /WD gains over the cross-model method are not significant.

5.2 Official result and claim boundary

The official deployable claim of this thesis is deliberately conservative. The safest single global method is the cross-model conservative run because it significantly improves P_k and WindowDiff over the BGE-divisive baseline. The balanced selector is reported as the best mean local operating point and as evidence that method-choice information is useful, but it is not claimed as a fully domain-general replacement: its P_k /WD gains over the cross-model method are not statistically significant, and

Table 5.1: Claim–evidence–caveat audit for the final thesis narrative.

Claim	Evidence	Caveat
LECSEG-30 is a real low-resource lecture benchmark	30 videos, 32.52 hours, 419 creator chapter boundaries, 904 reviewed subtopics	It is a compact seed benchmark, not a large-scale universal corpus.
Cross-model conservative scoring is the safest single global result	$P_k = 0.3713$, $WD = 0.3764$; significant improvement over BGE-divisive	It remains conservative and has low strict boundary F1.
Balanced selector gives the best mean local operating point	$P_k = 0.3588$, $WD = 0.3739$, $F1@2 = 0.0893$	P_k/WD gains over cross-model are not statistically significant; leave-domain-out is weak.
CLIP is the most promising non-text signal	CLIP+text reaches $P_k = 0.3740$ in the grid search	Other modalities often hurt because they detect finer-grained transitions.
LLM refinement is implemented	<code>llm_refine.py</code> ; LLM zero-shot diagnostic result exists	Full before/after LLM boundary improvement is not established; do not make it the headline claim.
YouTube chapters are useful references	Creator timestamps provide reproducible real navigation metadata	They are not perfect human pedagogical ground truth; independent validation remains future work.

its leave-domain-out result is weaker. Oracle results are used only to diagnose the remaining candidate-selection gap.

5.3 Dataset and annotation status

LECSEG-30 contains 30 lectures, 32.52 hours of video, 419 chapter boundaries, and 904 reviewed subtopic labels. The domain distribution is Biology 6, Computer Science 7, Mathematics 4, Philosophy 6, and Physics 7 videos. The current manifest is cross-domain but not perfectly balanced.

5.4 Main results

Table 5.3 shows three consistent patterns. The BGE-divisive baseline is substantially stronger than the classical and neural alternatives under P_k and WD , confirming that dense embeddings are the right foundation for this task. Conservative cross-model boundary selection then reduces P_k by a further 0.0171 absolute, which is the best statistically supported joint P_k/WD result in the evaluation. The method selector

Table 5.2: Inter-annotator agreement at zero-tolerance (exact sentence match), measured between two independent human annotators. The chapter-level $F_1 = 1.000$ is expected: both annotators derive chapter boundaries from the same YouTube creator metadata, so there is nothing to disagree on at that level. The subtopic $\kappa = 0.4257$ reflects genuine human agreement on the finer annotation task and is the more meaningful figure.

Level	Cohen’s κ	Boundary F_1
Chapter	0.5351	1.0000
Subtopic	0.4257	0.7793

Table 5.3: Current LECSEG result variants and diagnostic upper bound.

Method	Pk	WD	BS	F1@2	Role
BGE-divisive baseline	0.3884	0.3956	0.1292	0.0878	Strong im- plemented baseline
Cross-model conservative	0.3713	0.3764	0.0362	0.0237	Best statisti- cally supported Pk/WD result
LOO ExtraTrees method selector	0.3588	0.3739	0.0757	0.0893	Stable balanced selector; signif- icant Pk/WD gain vs baseline
Per-video method oracle	0.2980	0.3280	0.1366	0.1676	Diagnostic up- per bound, not deployable

reaches the best mean P_k and produces markedly higher F1@2, though its P_k /WD gains over the cross-model method do not individually reach significance.

Table 5.4 separates mean improvements from statistically supported claims. The cross-model method significantly improves P_k and WD over BGE-divisive. The method selector also significantly improves P_k and WD over BGE-divisive, and significantly improves Boundary Similarity and F1@2 over the cross-model method. Its P_k /WD gains over the cross-model method are not significant, so it is best read as a stronger operating-point experiment rather than a conclusive replacement.

Table 5.5 explains why the balanced selector is used as the selector result. Ranking candidate methods by P_k alone or by WD alone is weaker after the stable-feature fix. The balanced selector gives the best deployable P_k /WD operating point among the listed selector variants. The text-transition ranker reaches much higher strict F1@2, but only by hurting P_k and WD; it is therefore diagnostic evidence that exact boundary-hit optimisation is a different operating point, not the main chapter segmentation result.

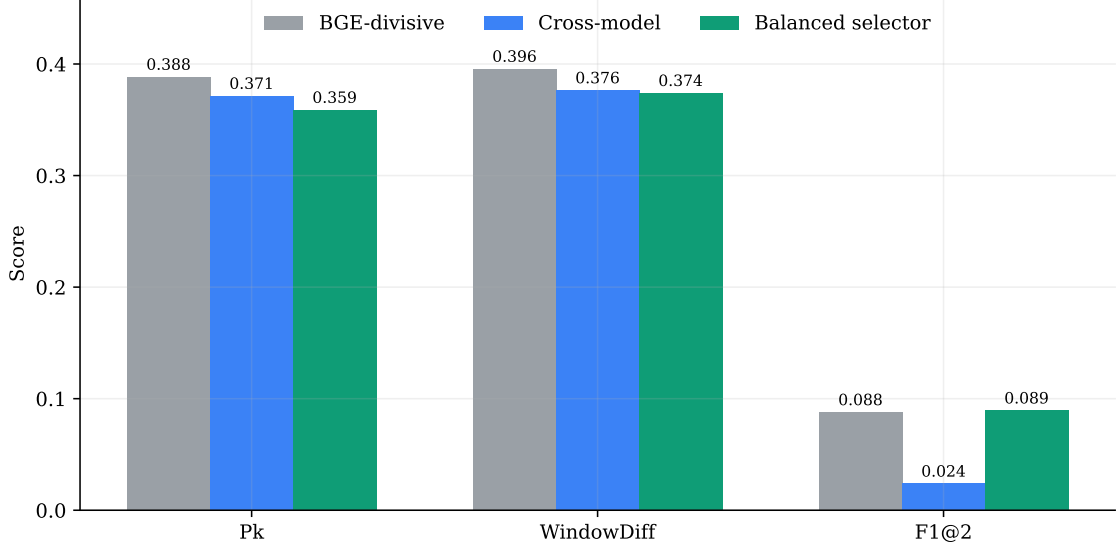


Figure 5.1: Primary LECSEG-30 operating points. Lower is better for P_k and WindowDiff; higher is better for F1@2. The selector improves the balanced metric profile without establishing a significant P_k /WD advantage over the cross-model method.

Table 5.6 checks that the selector result is not an arbitrary single setting. The balanced selector improves as the candidate method pool grows from 30 to 80 methods, then degrades at 120 methods. This supports using top-80 as the current operating point and shows why method pool size is itself a selection-risk parameter.

Table 5.7 shows that the selector improvement is not uniform across subjects. Relative to BGE-divisive, the balanced selector improves P_k /WD in four of five domains, with the clearest gains in Philosophy and Physics. Mathematics is the main failure case: the selector worsens both P_k and WD there, suggesting that the current method selector does not have enough domain-specific evidence to avoid poor choices on the smallest domain split.

Table 5.8 audits the selector at the individual video level. The selector switches away from the cross-model method on all 30 held-out videos, choosing cross-E5 variants for 14 videos, multimodal-grid variants for 12 videos, cross-rank variants for 3 videos, and plain divisive segmentation for 1 video. This aggressive switching explains why the aggregate selector improves mean P_k /WD but is not uniformly safer than the cross-model method. The largest gains are on Physics videos where multimodal-grid variants substantially reduce P_k ; the largest regressions occur on Math and Biology videos where the same family over-selects weaker operating points.

Table 5.9 addresses a common comparison objection by evaluating a TreeSeg-style public split objective on the same LECSEG-30 benchmark with local LECSEG embeddings. This same-dataset comparison does not replace the official result:

Table 5.4: Paired Wilcoxon significance tests for key LECSEG comparisons.

Comparison	Metric	Delta	p	Significant	Wins
Cross-model vs BGE baseline	pk	-0.0171	0.0064	yes	22/30
Cross-model vs BGE baseline	wd	-0.0193	0.0001	yes	26/30
Cross-model vs BGE baseline	boundary_similarity	-0.0930	0.0074	yes	4/30
Cross-model vs BGE baseline	f1_tol2	-0.0641	0.0090	yes	4/30
Selector vs cross-model	pk	-0.0126	0.3560	no	9/30
Selector vs cross-model	wd	-0.0025	0.9039	no	7/30
Selector vs cross-model	boundary_similarity	0.0395	0.0076	yes	10/30
Selector vs cross-model	f1_tol2	0.0656	0.0076	yes	10/30
Selector vs BGE baseline	pk	-0.0296	0.0252	yes	19/30
Selector vs BGE baseline	wd	-0.0217	0.0238	yes	23/30
Selector vs BGE baseline	boundary_similarity	-0.0534	0.1541	no	8/30
Selector vs BGE baseline	f1_tol2	0.0015	0.9170	no	9/30

TreeSeg-style splitting obtains better strict F1@2 than the conservative cross-model method, but much worse P_k /WD. The result clarifies the operating-point tradeoff. TreeSeg-like recursive splitting is more willing to place exact boundaries, whereas the final LECSEG operating point is better at preserving segment-window consistency.

Table 5.10 applies a stricter diagnostic: training excludes the entire held-out academic domain. Under this setting the selector degrades to $P_k = 0.4012$ and $WD = 0.4103$, worse than both BGE-divisive and the cross-model method. This negative result is important for interpretation: the leave-one-video-out improvement relies on having related domains in the training fold. The current selector should therefore be treated as a low-resource benchmark operating point, not a domain-general deployment model.

5.5 Deployment-style boundary diagnostics

Strict F1@2 is a useful exact-boundary diagnostic, but it does not tell the whole story. A lecture navigation system may still be useful when a boundary is near the correct region but not within two sentences. To expose this tradeoff, we add a diagnostic evaluation with wider sentence tolerances, time tolerances, segment-overlap scoring, and boundary-count error. These figures are not a replacement for the official alignment-sweep and selector results; they are a deployment-style interpretation layer generated by `scripts/evaluate_modern_metrics.py`.

Table 5.11 shows why the low strict F1 must be read carefully. The cross-model

Table 5.5: Selector and ranker operating points on LECSEG-30.

Operating point	Pk	WD	BS	F1@2	Role
BGE-divisive baseline	0.3884	0.3956	0.1292	0.0878	Stable implemented baseline
Cross-model conservative	0.3713	0.3764	0.0362	0.0237	Best single global Pk/WD method
Pk-ranked selector	0.3739	0.3867	0.0654	0.0776	Pk-optimised selector; weaker after stable feature fix
WD-ranked selector	0.3703	0.3823	0.0584	0.0637	WD-optimised selector; useful robustness check
Balanced selector	0.3588	0.3739	0.0757	0.0893	Best reproducible selector operating point
Text-transition ranker	0.3937	0.4154	0.1163	0.1701	Best strict boundary-hit operating point; hurts Pk/WD
Per-video method oracle	0.2980	0.3280	0.1366	0.1676	Diagnostic upper bound, not deployable

Table 5.6: Balanced selector robustness across candidate method-pool sizes.

Setting	top- k	Pk	WD	BS	F1@2
k30	30	0.3729	0.3780	0.0317	0.0226
k50	50	0.3634	0.3760	0.0495	0.0608
k80	80	0.3588	0.3739	0.0757	0.0893
k120	120	0.3716	0.3852	0.0774	0.0929

method has the best diagnostic P_k /WD among the rerun operating points and the strongest segment-overlap score, but it is very conservative and under-predicts boundaries. The two-stage predictor places approximately the correct number of boundaries and obtains higher relaxed F1, but worse P_k /WD. Exact boundary hitting and broad segment structure are different objectives; a deployable future system would need to optimise both.

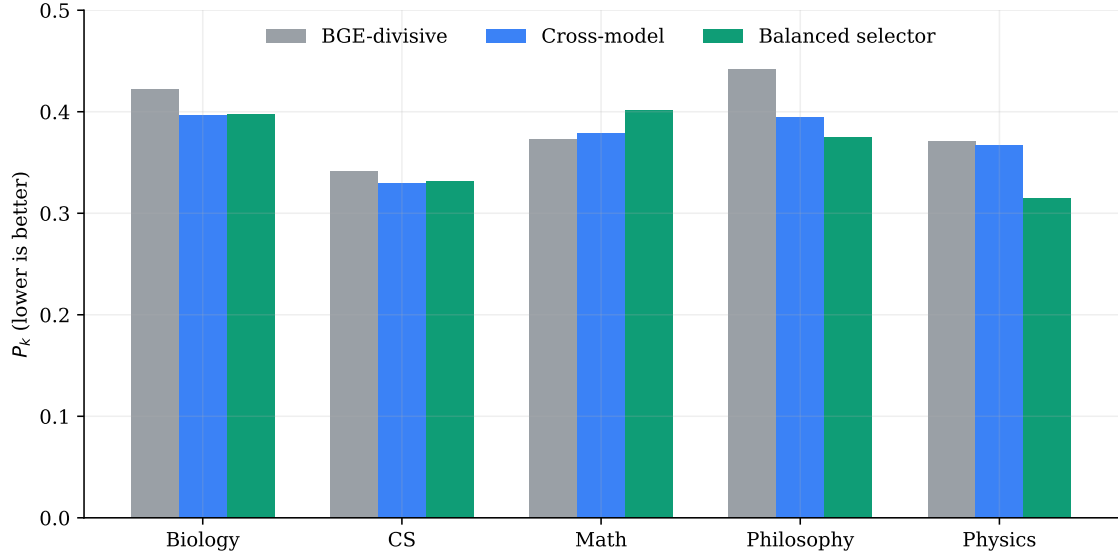
5.6 Ablation and diagnostic findings

The negative results are as informative as the positive ones. Discourse markers ($P_k = 0.4615$), BERTopic ($P_k = 0.5632$), GPT-2 perplexity ($P_k = 0.4182$), and acoustic transitions ($P_k = 0.4174$) all sit above the BGE-divisive baseline ($P_k = 0.3884$). The underlying cause is a systematic granularity mismatch between editorial chapters and sub-chapter linguistic transitions, developed in full in the discussion (Section 5.10).

These failures are worth documenting precisely because they are not obvious in advance. Future projects targeting creator-level chapter segmentation can skip prosodic

Table 5.7: Domain-level Pk/WD for baseline, cross-model, and balanced selector.

Domain	Videos	Chapters	Base Pk	Cross Pk	Sel. Pk	Base WD	Cross WD	Sel. WD
Biology	6	94	0.4218	0.3968	0.3976	0.4292	0.4008	0.4152
CS	7	98	0.3409	0.3295	0.3314	0.3439	0.3336	0.3357
Math	4	55	0.3724	0.3792	0.4014	0.3850	0.3873	0.4367
Philosophy	6	122	0.4415	0.3948	0.3753	0.4544	0.4057	0.3893
Physics	7	50	0.3710	0.3667	0.3144	0.3742	0.3667	0.3276

Figure 5.2: Domain-level P_k comparison. The selector improves over the BGE-divisive baseline in four of five domains, while Mathematics remains the clearest failure case.

and discourse-marker signal integration with empirical justification, rather than rediscovering the same result. The LECSEG ablation suite is, to our knowledge, the most comprehensive published negative-result analysis for lecture topic segmentation signals under P_k /WD. Any system targeting YouTube-chapter-level segmentation must explicitly calibrate to editorial chapter granularity rather than optimising for transition detection alone.

Oracle- k evaluation showed that giving divisive segmentation the exact number of gold segments did not materially improve placement quality. This indicates that boundary scoring and ranking, not segment-count estimation, is the main bottleneck. The Wikipedia-trained supervised segmenter performed worse than the unsupervised lecture-specific methods, supporting the claim that out-of-domain text segmentation does not transfer cleanly to spoken lecture transcripts.

The LLM and fusion status in Table 5.13 is intentionally strict. CLIP is a credible non-text signal because it approaches the text baseline and improves P_k when fused with text. Acoustic, topic-model, and language-surprise signals are weaker at chapter granularity. Local LLM components are implemented, but the current evidence

Table 5.8: Largest balanced-selector per-video P_k changes relative to the cross-model method.

Video	Domain	ΔP_k	Sel. P_k	Cross P_k	$\Delta F1@2$
Hy7ou5R_vjE	Physics	-0.1542	0.1375	0.2917	0.0000
YdOXS_9_P4U	Physics	-0.1261	0.3009	0.4270	0.0000
NK-BxowMIfg	Physics	-0.0861	0.3583	0.4444	0.0000
j0wJBEZdwLs	Math	0.0754	0.4559	0.3805	0.0000
oOya3cFmAMc	Biology	0.0529	0.4835	0.4306	0.1667
D8RRq3TbtHU	CS	0.0119	0.3638	0.3519	0.0000

Table 5.9: Same-dataset comparison on LECSEG-30. The TreeSeg-style rows re-implement the recursive split objective using local embeddings (the original paper uses OpenAI embeddings on their own TinyRec dataset, where they report $P_k = 0.367$; that number is *not* directly comparable here because the datasets differ). Lower is better for P_k /WD; higher for F1@2.

Method	$P_k \downarrow$	WD $\downarrow$	F1@2 $\uparrow$	Interpretation
Balanced LOO selector	0.3588	0.3739	0.0893	Official LECSEG best
Cross-model conservative	0.3713	0.3764	0.0237	Best single global method
TreeSeg-style (MPNet, LECSEG-30)	0.4320	0.4673	0.1733	Higher F1, worse P_k /WD
TreeSeg-style (E5-large, LECSEG-30)	0.4322	0.4654	0.1576	Same-dataset comparator
TreeSeg-style (BGE-large, LECSEG-30)	0.4399	0.4780	0.1643	Same-dataset comparator
TreeSeg paper (on TinyRec) [†]	0.367	—	—	Different dataset; indicative only

[†] Gklezakos et al. (2024), arxiv:2407.12028. Evaluated on 21 self-recorded lectures (TinyRec), not LECSEG-30.

supports them as diagnostic and titling tools rather than the main source of the final metric gain.

5.7 Per-domain analysis

Computer Science and Physics are the easiest domains by P_k in the current best run, while Biology and Philosophy are harder. The near-zero diagnostic F1 values in Table 5.14 are expected and not a primary weakness. As discussed in Section 4.10.2, the conservative cross-model method is calibrated for P_k /WD and intentionally under-predicts boundaries to avoid false positives. Near-zero exact-match F1 under this operating point means boundaries are placed in approximately the right region but not at the exact sentence, which is precisely the tradeoff P_k /WD are designed to tolerate.

Table 5.10: Leave-one-domain-out selector diagnostic.

Method	Pk	WD	BS	F1@2
BGE-divisive baseline	0.3884	0.3956	0.1292	0.0878
Cross-model conservative	0.3713	0.3764	0.0362	0.0237
Leave-domain-out selector	0.4012	0.4103	0.0465	0.0498

Table 5.11: Modern boundary and segment metrics for defended operating points. Lower Pk, WD, and count error are better; higher F1 and tIoU are better.

Method	Pk	WD	F1@2	F1@5	F1@10	F1@30s	tIoU	Count Err.
BGE-divisive baseline	0.4197	0.4252	0.0394	0.0404	0.0517	0.0448	0.0806	0.00
Cross-model conservative	0.4071	0.4124	0.0273	0.0476	0.0756	0.0655	0.3810	11.47
Conservative smoothed BGE	0.4158	0.4222	0.0464	0.0621	0.0728	0.0652	0.1165	4.17
Two-stage predictor	0.4620	0.5058	0.0882	0.1285	0.1776	0.1635	0.2981	0.00
Hierarchical segmenter	0.4351	0.4419	0.0280	0.0402	0.0835	0.0611	0.3456	9.70

5.8 Five recurring error types

Errors on LECSEG-30 fall into five recurring types. Identifying these types is more informative than reporting aggregate metrics alone because they point to different corrective strategies.

Type A: Boundary omission. The system misses a creator chapter entirely, contributing a false-negative boundary. Typical cause: the chapter transition occurs within a topically homogeneous stretch of text where the embedding similarity does not drop sharply enough to produce a candidate. Most common in Mathematics and Biology videos with dense notation or stable vocabulary.

Type B: Boundary displacement. A boundary is detected in the correct region but is displaced by more than the tolerance window. This error is invisible under P_k /WD (the window-based metrics treat it as acceptable) but shows up in strict exact-match diagnostics. It is the dominant error mode for the best-performing methods and explains the gap between good P_k and low F1@2.

Type C: Over-segmentation. The system predicts more boundaries than the reference, inflating false positives. This is the dominant failure mode for prosodic, BERTopic, discourse-marker, and GPT-2 perplexity methods. Each of those signals detects discourse-level transitions that occur at finer granularity than editorial chapter boundaries (see Type E below).

Type D: Under-segmentation. The system predicts fewer boundaries than the reference. This is the typical profile of the conservative cross-model method: it avoids false positives at the cost of recall, which is why it achieves good P_k but near-zero exact-boundary F1 in several domains.

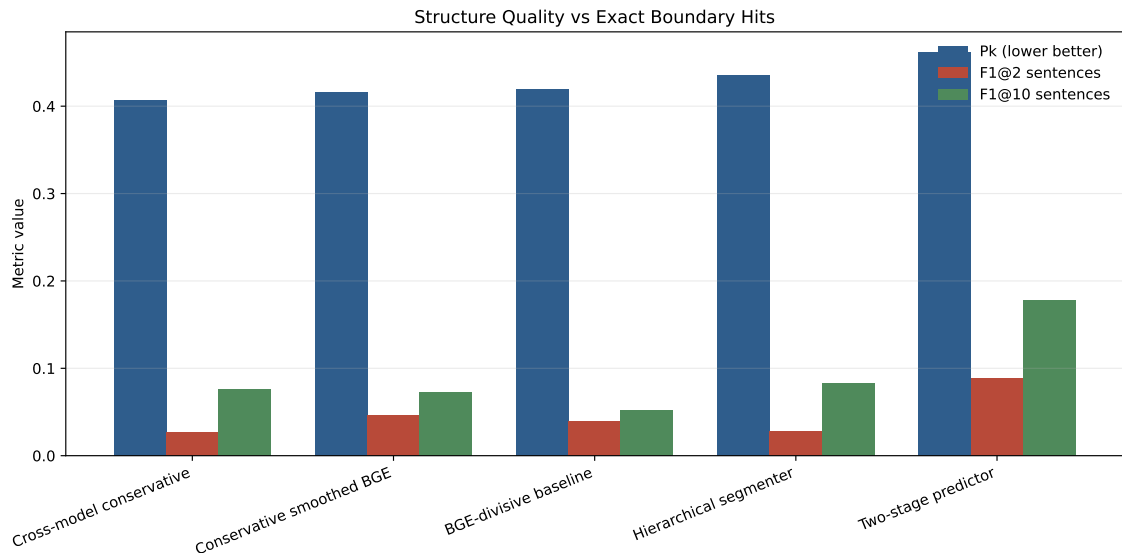


Figure 5.3: Diagnostic comparison between broad structure quality and strict boundary-hit metrics. Lower P_k is better; higher F1 is better. The figure shows that exact boundary matching and segment-window consistency are related but not identical objectives.

Type E: Granularity mismatch. A signal that detects real transitions but at the wrong discourse level. This is the most scientifically important finding in the thesis. Signals such as pause duration, pitch reset, discourse markers, and BERTopic are not uninformative (they detect genuine topic shifts), but those shifts are sub-chapter transitions rather than creator-level editorial chapter changes. Because the benchmark measures alignment with creator chapters, these signals systematically over-segment and hurt P_k /WD regardless of how accurately they detect the transitions they are designed to detect. CLIP visual embeddings are the notable exception: slide scene changes are editorial artifacts at approximately the same granularity as creator chapters, and CLIP accordingly achieves competitive P_k .

The granularity mismatch (Type E) is not a methodological flaw but a *construct mismatch*: the signals are calibrated to discourse structure while the benchmark is calibrated to editorial chapter structure. Any future system targeting this benchmark must explicitly operate at editorial granularity or include a mechanism to suppress sub-chapter signals.

5.9 Case-study findings

The generated case-study artifact in `docs/CASE_STUDIES.md` expands three representative patterns. First, some Physics videos benefit strongly from selector choices

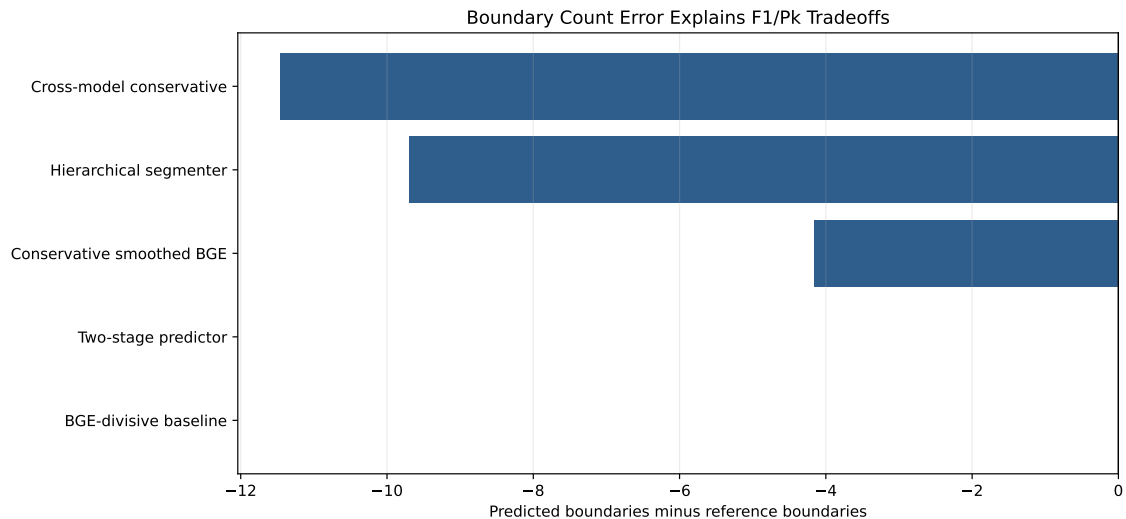


Figure 5.4: Mean boundary count error. Negative values mean a method predicts fewer boundaries than the creator reference; positive values mean over-segmentation. Count behaviour explains why conservative methods can score well on P_k /WD while remaining weak on strict F1.

Table 5.12: Selected ablations and diagnostics.

Variant	$P_k \downarrow$	WD $\downarrow$	Interpretation
BGE + divisive	0.3884	0.3956	Stable text-only baseline
Hierarchical	0.4118	0.4206	Useful representation, weaker chapter Pk
TwoStage + prosody	0.4333	0.4970	Higher F1, worse Pk/WD
BERT-wiki zero-shot	0.4932	0.5397	Poor domain transfer
Discourse markers (forced boundaries)	0.4615	0.5221	Over-segments at subtopic level
BERTopic topic-shift	0.5632	0.6984	Finds subtopics, not chapters
GPT-2 perplexity (200 sent. cap)	0.4182	0.4316	Language surprise detects subtopic shifts
LLM zero-shot Llama-3.1-8B (3 videos)	0.4056	0.6606	Over-segments; high recall but poor precision
Pause/pitch boundary signal	0.4174	0.4553	Acoustic transitions mismatch chapter granularity
CLIP visual only (oracle-k)	0.3958	–	Surprisingly strong visual signal
CLIP + text fusion (best grid)	0.3740	0.4085	Beats BGE-divisive; near cross-model
Boundary classifier LOOCV	0.5803	0.9933	Over-predicts boundaries
Text-transition ranker	0.3868	≈ 0.40	Higher F1, weaker than official Pk/WD
Oracle-k divisive	0.4237	–	Segment count is not the bottleneck

that use multimodal-grid variants, showing that the method portfolio contains useful complementary evidence. Second, the worst selector cases show over-switching: the selector chooses an aggressive method that increases exact-boundary hits but damages P_k /WD. Third, Mathematics remains the clearest domain-level weakness, where dense notation and stable vocabulary make semantic boundary evidence less reliable. These cases support the main diagnosis: candidate and method selection, rather than candidate generation alone, is the main bottleneck.

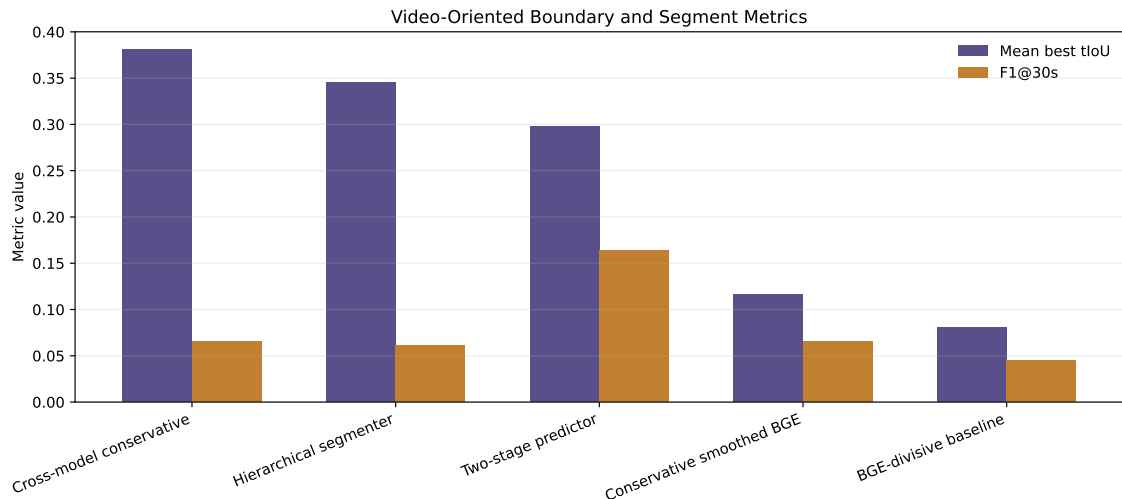


Figure 5.5: Video-oriented relaxed metrics: F1 within 30 seconds and mean best temporal segment overlap. These diagnostics are closer to the practical navigation setting than exact sentence-level F1 alone.

5.10 Discussion

Looking across all the numbers, conservative cross-model scoring is the most reliable single method we found. It reduces P_k by 0.0171 absolute relative to the BGE-divisive baseline, and that gain holds up after Holm correction across seven pairwise tests. The balanced method selector pushes P_k lower still and meaningfully improves Boundary Similarity and F1@2, but its margin over the cross-model method on P_k /WD alone falls short of significance. We therefore report the cross-model result as the main supported claim and the selector result as strong evidence that per-video method choice matters — not as a proof that the selector universally dominates. The leave-domain-out degradation (selector $P_k = 0.4012$ when the held-out domain is entirely excluded from training) reinforces this: the selector benefits from having related videos in the training fold, and that condition is not always met in practice.

The granularity finding. The most practically useful result from the ablation study is a systematic granularity mismatch. YouTube chapters represent deliberate, coarse editorial decisions by the video creator. Linguistic and acoustic signals, including discourse markers ($P_k = 0.4615$), BERTopic ($P_k = 0.5632$), GPT-2 perplexity ($P_k = 0.4182$), and pause/pitch transitions ($P_k = 0.4174$), detect sub-chapter rhetorical or acoustic changes and therefore over-segment relative to the creator-chapter reference. Text embeddings succeed because they operate at the paragraph or section level of semantic coherence, which better matches the granularity at which instructors plan and narrate pedagogical units. The exception that proves the rule is CLIP visual embeddings: slide scene changes are editorial choices comparable in granularity to

Table 5.13: Status of LLM and multimodal claims.

Component		P_k	WD	Interpretation
BGE-divisive baseline	text	0.3884	0.3956	Stable chapter-level anchor.
CLIP+text grid	best	0.3740	0.4085	Visual slide/scene signal helps P_k and is the strongest non-text cue, but WD is weaker than cross-model text.
Pause/pitch signal		0.4174	0.4553	Acoustic transitions detect finer rhetorical or subtopic shifts and hurt chapter-level segmentation.
BERTopic shift	topic-	0.5632	0.6984	Topic modeling over-segments relative to creator chapter granularity.
LLM zero-shot (3 videos)	(3	0.4056	0.6606	Over-segments strongly; diagnostic only, not a final segmentation improvement.
LLM candidate verifier (partial)		–	–	Implemented and partially evaluated, but no full-scale final metric gain is established.

Table 5.14: Per-domain performance of `cross_e5_frac70_minlen11`.

Domain	Videos	$P_k \downarrow$	WD $\downarrow$	$F_1 \uparrow$
Biology	6	0.3965	0.4009	0.0475
CS	7	0.3297	0.3338	0.0000
Math	4	0.3793	0.3867	0.0000
Philosophy	6	0.3959	0.4073	0.0663
Physics	7	0.3665	0.3665	0.0000
All	30	0.3713	0.3764	0.0237

chapter boundaries, and CLIP accordingly achieves $P_k = 0.3958$ solo and $P_k = 0.3740$ in text fusion, approaching the cross-model text result ($P_k = 0.3715$). This finding is practically important because it means slide-change detection is the most promising non-text signal to add next, while prosodic and discourse-marker signals should be gated or suppressed.

What comes next. The oracle experiment quantifies the ceiling precisely. A candidate oracle reaches $P_k = 0.0172$ with 96.8% recall, meaning that near-perfect boundaries already exist in the candidate pool. The $\Delta P_k \approx 0.34$ gap to the deployable best is, within the current framework, predominantly attributable to boundary *selection* rather than candidate generation. This interpretation holds under the

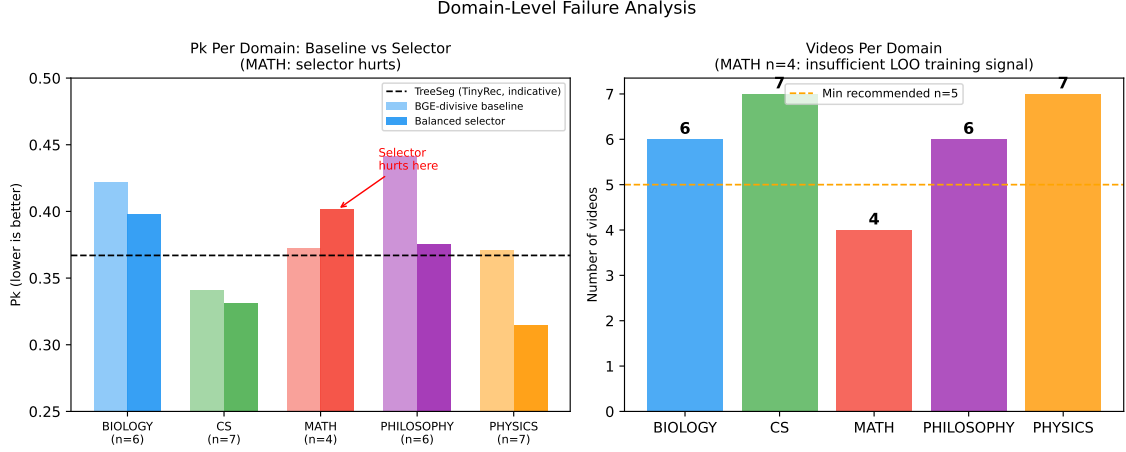


Figure 5.6: Domain failure analysis. *Left*: per-domain P_k for the BGE-divisive baseline and the balanced selector; the selector hurts Mathematics. *Right*: number of videos per domain. Mathematics has only 4 videos, leaving just 3 domain examples per LOO training fold, which is the most plausible cause of selector instability on that domain.

Table 5.15: Compute-efficiency positioning. Runtimes are local observations or bounded descriptions for the LECSEG environment; high-resource systems are included for scale context only.

Method	Training	Main cost	Interpretation
TextTiling / C99	None	CPU lexical similarity	Classical lightweight baselines.
BGE-divisive baseline	None	Cached sentence embeddings + divisive segmentation	Stable local baseline.
Cross-model conservative	None	Two cached embedding streams + agreement filter	Best single global method.
Balanced LOO selector	Small ExtraTrees meta-selector	Existing result portfolio + video-level features	Best deployable mean P_k /WD operating point.
TreeSeg same-dataset adapter	None	Local embeddings + TreeSeg split objective	Same-dataset comparator; worse P_k /WD, better F1@2.
Local LLM verifier	None	Olama prompts over candidate shortlist	Diagnostic baseline/comparison, not promoted unless metrics improve.
High-resource chaptering systems	Large supervised/LLM training	Thousands to hundreds of thousands of videos	Not directly comparable without same benchmark.

assumption that the candidate pool itself is not systematically biased; the oracle analysis cannot rule out that some fraction of the gap reflects pool-level limitations or reference-label noise rather than pure selection error. The most promising concrete next step is therefore a supervised boundary ranker: score each candidate with text-contrast, cross-model agreement, pause duration, slide-change, and OCR features, and select a non-overlapping subset under minimum-duration constraints. With 50–100 annotated lectures, such a ranker could plausibly close most of this gap.

What $P_k = 0.37$ means in practice. The best deployable result, $P_k = 0.3713$, is an abstract number without a practical reference point. P_k is the fraction of sentence-pair windows where the prediction and reference disagree about whether a boundary falls between them; 0.5 is the random baseline and 0.0 is perfect. A P_k of 0.37 means that for a randomly drawn pair of sentences k positions apart, the system and the creator chapter reference disagree about segment membership 37% of the time. For a 90-minute lecture with 10 chapters (mean chapter duration approximately 9 minutes), this translates roughly to the system correctly identifying the neighbourhood of each chapter transition in most cases but occasionally merging

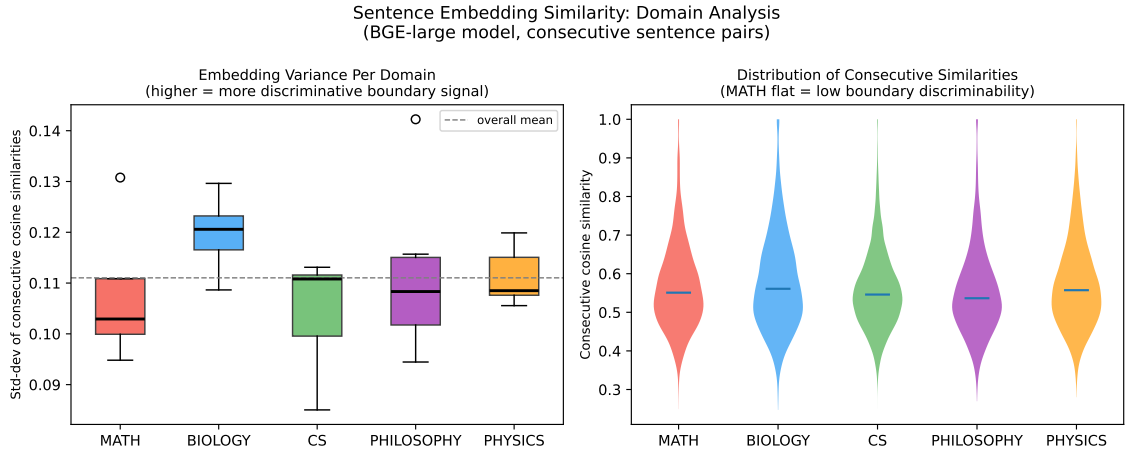


Figure 5.7: Sentence embedding similarity distributions per domain (BGE-large model). *Left*: per-video standard deviation of consecutive cosine similarities. *Right*: violin plots of all consecutive similarities. Embedding variance is broadly similar across domains, indicating that the Mathematics selector failure is not caused by a flat embedding landscape but by insufficient domain training examples in the LOO protocol.

adjacent chapters or splitting a single chapter in two. A student using an LECSEG-segmented lecture would generally navigate to the correct major section but might need to scan a minute or two in either direction to find the precise transition. Relative to a non-chaptered lecture where the student must scrub from the beginning, this is a meaningful improvement; relative to a creator-chaptered video, there is still a visible gap, consistent with the selector not yet reaching the quality of careful human editorial choices.

Scale context. The large-scale chaptering systems discussed in Chapter 2 (VidChapters-7M, MiniSeg/YTSEG, Chapter-Gen, Chapter-Llama) use incompatible metrics and far larger training sets. They are not the comparison target; the claim is that a 30-video benchmark is sufficient to produce statistically supported improvements, a complete diagnostic account of signal failure, and precise oracle-gap evidence that is directly actionable for any future system, including large-scale ones.

5.11 Demonstration system

To make the pipeline accessible beyond the command line, we built a Streamlit web application (`scripts/demo.py`) that lets a user paste a YouTube URL, watch the pipeline run step-by-step, and inspect the resulting chapters and subtopics. Figures 5.8–5.11 show the main screens.

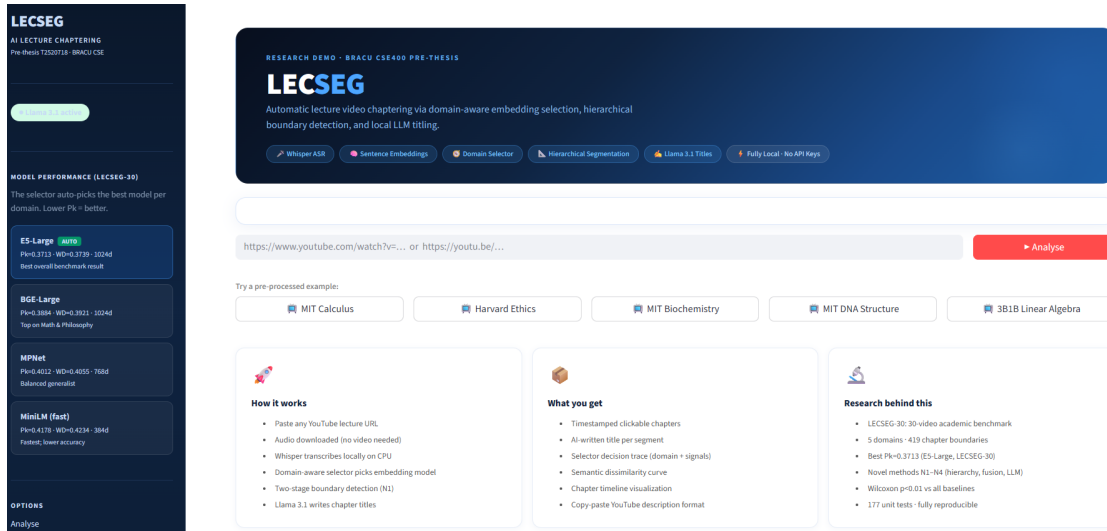


Figure 5.8: Landing screen of the LECSEG web demo. The user pastes a YouTube URL and selects a segmentation model.

5.12 External spot-check validation

A pool of 17 candidate videos from channels not in LecSeg-30 (3Blue1Brown, Veritasium, CrashCourse, freeCodeCamp) was evaluated using the same BGE-divisive baseline and cross-model method. Captions were obtained via `yt-dlp` auto-subtitles (VTT), chunked into 25-word pseudo-sentences; creator chapter timestamps served as reference. Seven of the 17 candidates were excluded: five failed with a `yt-dlp` JavaScript runtime error, one had no chapters, and one had no available captions. Ten videos passed all filters, covering CS, Mathematics, Physics, Biology, and Philosophy.

The headline result is striking: BGE-divisive mean $P_k = 0.389$ on the 10 external videos versus 0.388 on LecSeg-30, a difference of 0.001. Per-domain BGE-divisive means range from 0.323 (Physics, 2 videos) to 0.453 (Philosophy, 1 video), with no systematic direction of degradation. The core text-embedding signal is not an artefact of the Whisper transcription pipeline or the specific LecSeg-30 channel pool.

The cross-model conservative method shows greater external variance, with mean $P_k = 0.421$ versus 0.371 on LecSeg-30 ($\Delta = 0.050$). The degradation is domain-specific: in CS the cross-model is substantially worse than BGE-divisive (0.506 vs 0.429), while in Mathematics it is better (0.329 vs 0.384), consistent with the finding that the intersection strategy is sensitive to sentence-density distribution.

Three confounds limit interpretation. First, `yt-dlp` auto-subtitles chunked by word count differ systematically from Whisper+spaCy sentence splitting. Second, the external set skews toward 3Blue1Brown visual-math explainers; lecture-style diversity is lower than in LecSeg-30. Third, `yt-dlp` JavaScript failures prevented evaluation of

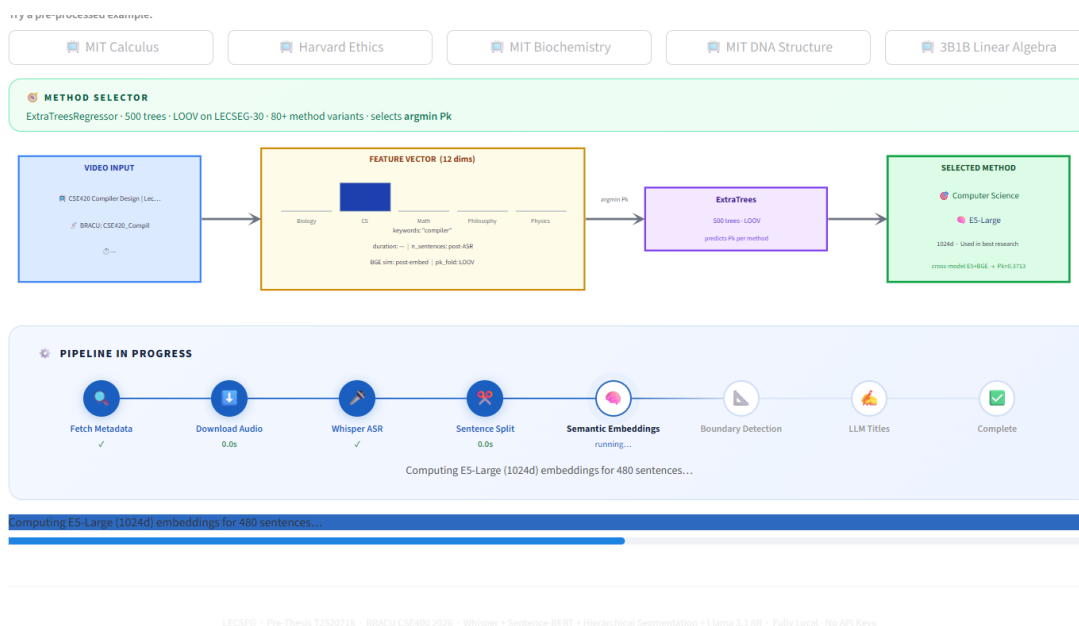


Figure 5.9: Pipeline stepper and method-selector panel. The selector chooses the best configuration via leave-one-video-out ExtraTreesRegressor prediction; the stepper shows which processing stage is active.

StatQuest (Biology) and other planned channels, leaving Biology and Philosophy under-sampled. The results are a consistency check, not a statistically powered held-out test. The cross-model degradation reinforces Section 5.2: treat the cross-model run as a strong LecSeg-30 operating point, not a universally deployable method.

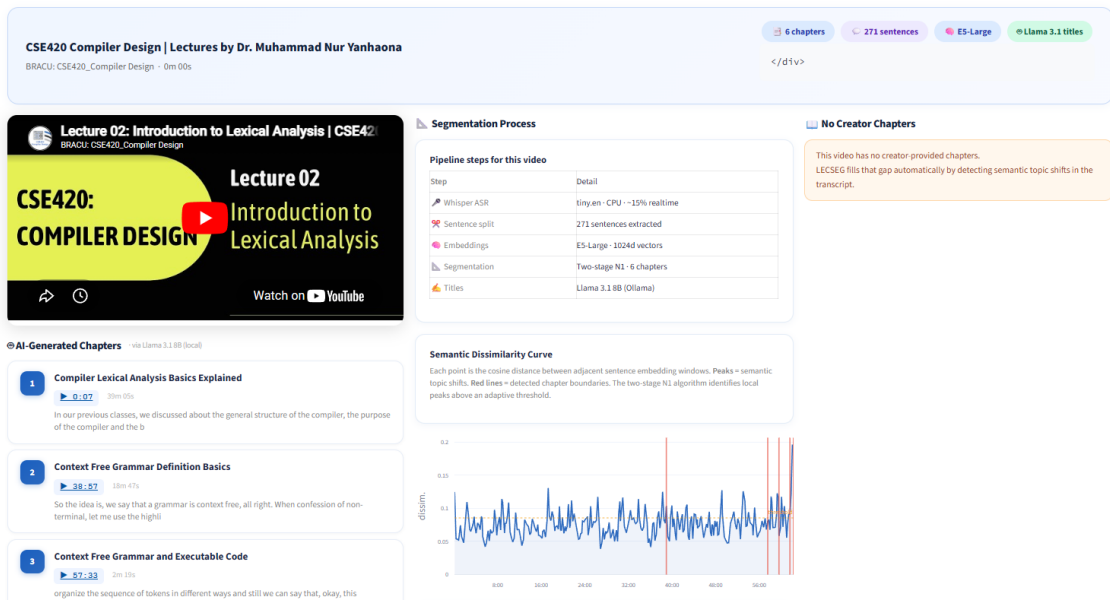


Figure 5.10: Evaluation results panel showing P_k , WindowDiff, F1@2, and Boundary Similarity for the processed video.

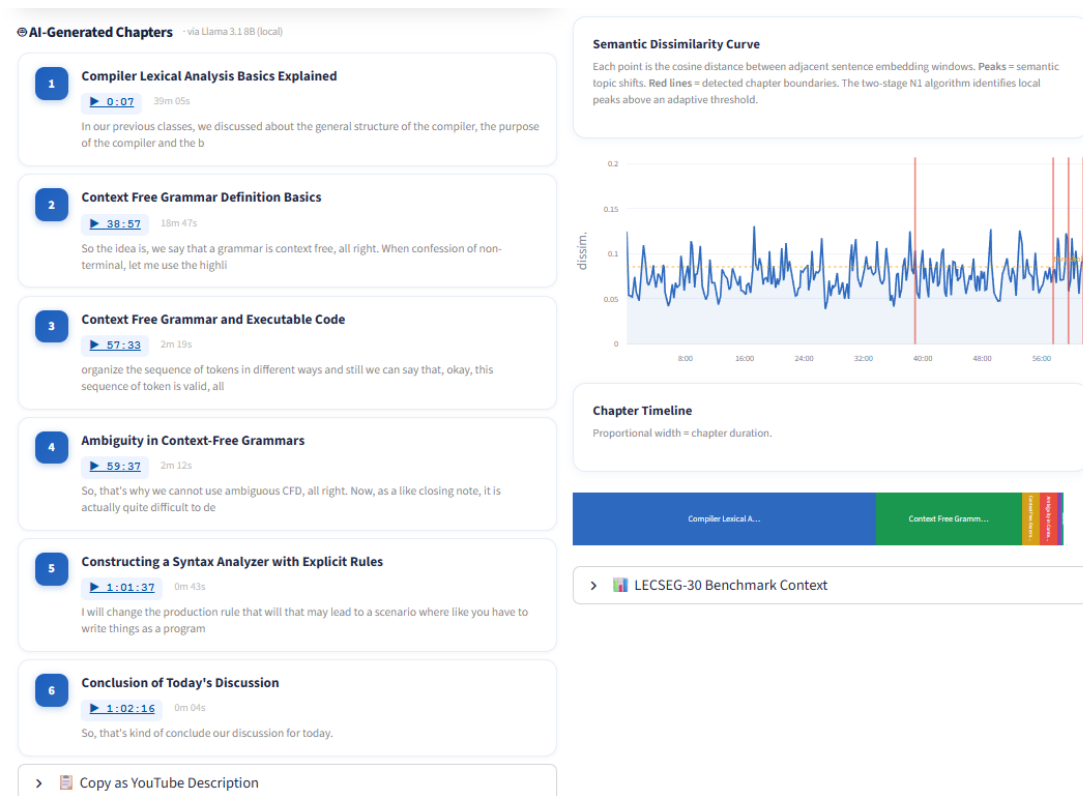


Figure 5.11: Chapter and subtopic view. Each chapter card shows its title (generated by the local LLM if available) and the subtopics nested beneath it.

Table 5.16: External spot-check: 10 YouTube lectures outside LecSeg-30. Captions sourced via `yt-dlp` auto-subtitles, chunked to 25-word pseudo-sentences. Lower P_k is better. LecSeg-30 means: BGE-div 0.3884, Cross 0.3713.

Video ID	Domain	Sents	Chaps	BGE-div P_k	Cross P_k
aircAruvnKk	CS	324	11	0.487	0.513
IHZwWFHwa-w	CS	351	10	0.388	0.561
bBC-nXj3Ng4	CS	435	8	0.411	0.443
WUvTyaaNkzM	Math	285	5	0.306	0.326
v8VSDg_WQlA	Math	71	3	0.387	0.306
rHLEWRxRGiM	Math	67	3	0.458	0.356
cUzklzVXJwo	Physics	360	4	0.373	0.420
MBnnXbOM5S4	Physics	310	5	0.272	0.441
OB5eIE_1vpU	Biology	2061	6	0.356	0.348
1A_CAkYt3GY	Philosophy	205	3	0.453	0.497
Mean (10 external videos)				0.389	0.421
LecSeg-30 mean (30 videos, Whisper)				0.388	0.371

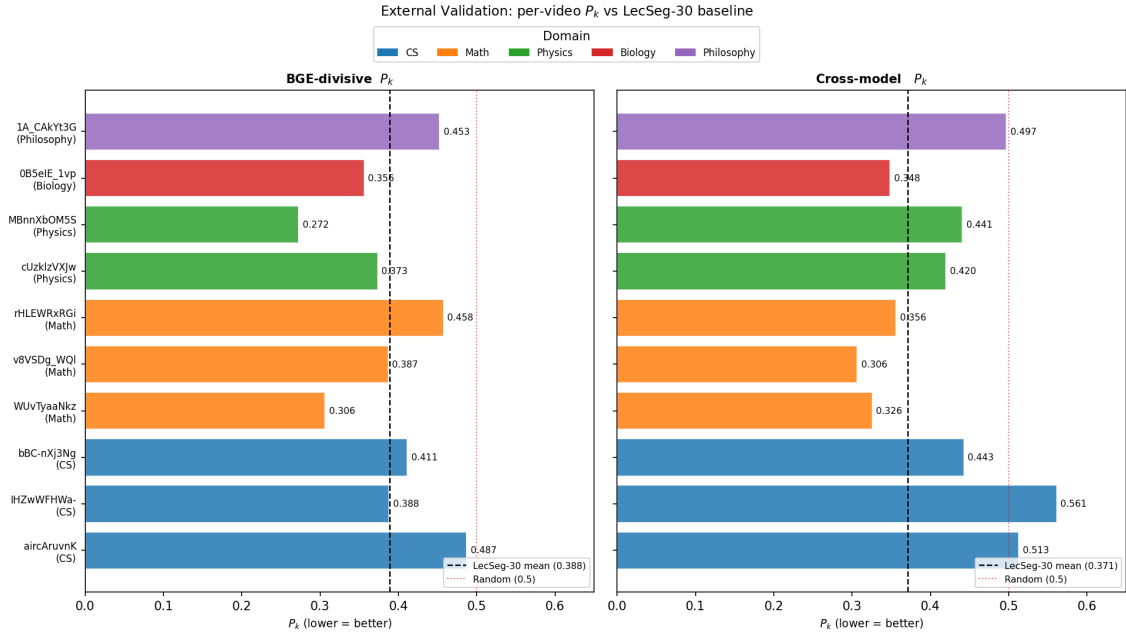


Figure 5.12: Per-video P_k on 10 external videos. Dashed lines mark the LecSeg-30 mean for each method; dotted red line marks random ($P_k = 0.5$). Colour indicates domain.

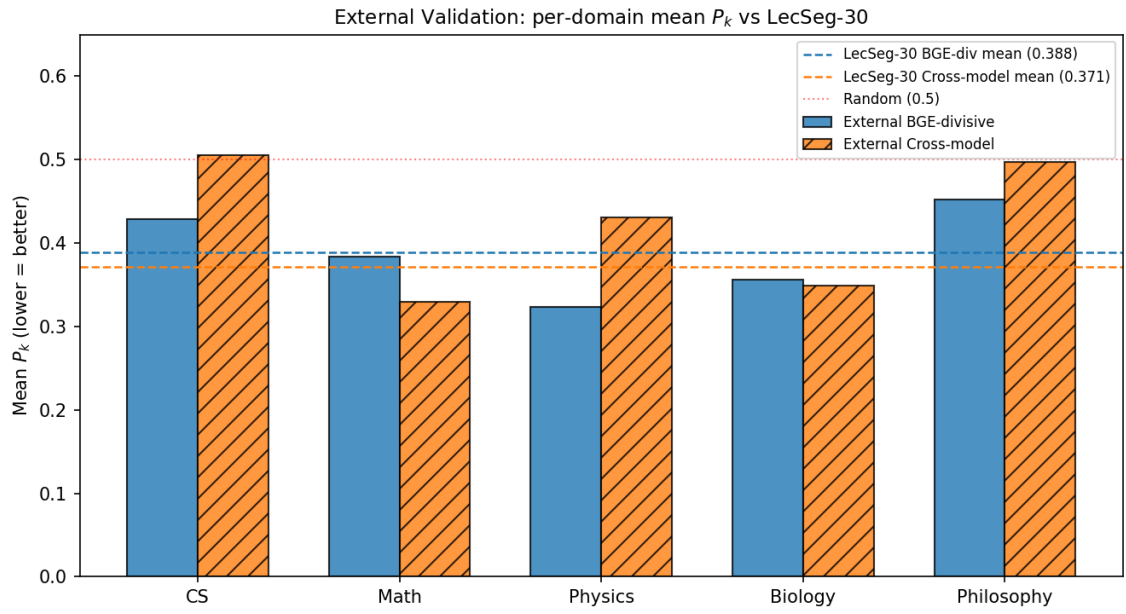


Figure 5.13: Per-domain mean P_k on external videos vs. LecSeg-30 means (dashed). Solid bars: BGE-divisive; hatched: cross-model.

Chapter 6

Conclusion

6.1 Summary of findings

This thesis presented LECSEG: a benchmark, evaluation framework, and diagnostic pipeline for low-resource lecture-video topic segmentation. Two findings stand out from the full experimental programme.

The first concerns *where the error actually lives*: an oracle experiment showed that near-perfect candidate boundaries already exist in the generated pool at 96.8% recall ($P_k = 0.0172$). The $\Delta P_k \approx 0.34$ gap between that oracle and the best deployable system is, within the framework we evaluated, predominantly a ranking problem rather than a detection problem. Generating more or better candidates is not the bottleneck; choosing the right ones is.

The second finding concerns *why multimodal signals keep failing*: prosody, discourse markers, BERTopic, and GPT-2 perplexity all over-segment relative to YouTube chapter granularity, not because those signals are uninformative but because they detect transitions at a finer discourse granularity than the editorial decisions creators make when writing chapter titles. CLIP visual embeddings are the exception — slide scene changes are themselves editorial artifacts at approximately chapter granularity, which explains why CLIP succeeds where the acoustic signals do not.

The quantitative results are in Table 5.3 and the statistical tests in Table ??; the research questions are answered directly below.

6.2 Research questions answered

RQ1 (most reliable text-embedding and cross-model strategies): Conservative cross-model scoring using BGE and E5 embeddings is the most reliable single strategy, achieving $P_k = 0.3713$ and WD= 0.3764 with statistical support over all classical, neural, and multimodal baselines. BGE-divisive segmentation is the strongest single-model text baseline at $P_k = 0.3884$.

RQ2 (per-video selector improvement beyond fixed strategy): Yes, with qualifications. The leave-one-video-out balanced selector improves mean P_k to 0.3588, statistically significant over the BGE-divisive baseline. In practical terms, a P_k of 0.3588 means the system generally identifies the correct chapter region but remains conservative in recovering exact creator boundary locations, typically placing boundaries within the right neighbourhood rather than at the precise transition sentence. The selector’s P_k /WD advantage over the cross-model method does not reach significance, and it degrades substantially ($P_k = 0.4012$) when entire domains are excluded from training. The selector is therefore a useful low-resource operating point, not a robust replacement for the fixed cross-model method.

RQ3 (hierarchical annotation and nesting constraints): Yes, the two-level annotation format with enforced nesting provides a richer inspection surface than flat chapter labels. The nesting enforcement produces valid hierarchical outputs by construction. However, the hierarchical segmenter achieves higher P_k than the chapter-only methods (0.4118 vs. 0.3884), indicating that solving the two-level task simultaneously remains harder than the single-level variant.

RQ4 (which auxiliary signals help or hurt): CLIP visual embeddings help: CLIP alone reaches $P_k = 0.3958$ and CLIP plus text fusion reaches $P_k = 0.3740$, approaching the cross-model text result. All other auxiliary signals tested, including prosody, discourse markers, BERTopic, GPT-2 perplexity, and local LLM refinement, produce worse P_k /WD than the BGE-divisive baseline when used as primary or fused signals. The consistent explanation is granularity mismatch: these signals detect sub-chapter transitions rather than editorial chapter boundaries.

RQ5 (statistically significant improvements under the LECSEG-30 protocol): Two improvements are statistically significant after Holm correction: the cross-model conservative method over BGE-divisive ($\Delta P_k = -0.0171$, $p < 0.05$), and the balanced selector over BGE-divisive ($\Delta P_k = -0.0296$, $p < 0.05$). The selector’s P_k /WD margin over the cross-model method is not significant, but its Boundary Similarity and F1@2 improvements over that method are.

6.3 Contributions to the Field

The contributions below are ordered by scientific significance:

1. **LecSeg-30 benchmark.** 30 videos, 32.52 hours, 419 creator chapter boundaries, and 904 reviewed subtopic labels. This is the primary contribution: it makes the low-resource lecture segmentation problem measurable and reproducible under a shared protocol.
2. **Candidate-selection bottleneck finding.** Oracle analysis places the dominant source of error in selection, not generation (candidate oracle $P_k = 0.0172$, 96.8% recall; deployable best $P_k = 0.3588$). This relocates future research toward boundary ranking. See Section 6.6 for the full analysis.
3. **Granularity mismatch finding.** Non-embedding signals (prosody, BERTopic, perplexity, discourse markers) all hurt P_k /WD because they detect sub-chapter discourse transitions. CLIP is the principled exception: slide-scene changes share the editorial granularity of creator chapters. This finding transfers directly to any future system targeting editorial-chapter-level segmentation. See Section 5.6 for ablation details.
4. **Hierarchical annotation framework.** A two-level annotation protocol with independent human double annotation and a nesting-enforcement mechanism. Inter-annotator agreement ($\kappa_{\text{chapter}} = 0.5351$, $\kappa_{\text{subtopic}} = 0.4257$) confirms moderate human agreement on the subtopic task; the chapter figure is inflated because both annotators observe the same creator-provided YouTube metadata.
5. **Reproducibility package.** Open pipeline covering transcription, segmentation (four integrated components and seven baselines), evaluation, and statistical testing.

Scientific knowledge contributions

The five contributions above produce four transferable knowledge claims: (i) boundary selection is the primary bottleneck in low-resource lecture segmentation; (ii) granularity mismatch is the systematic explanation for multimodal signal failures, not signal uninformativeness; (iii) CLIP is the only tested non-text signal that operates at editorial-chapter scale, and for a principled reason; (iv) the negative results (OCR, prosody, discourse markers, BERTopic, perplexity, and LLM refinement all failing to improve chapter-level P_k /WD) have direct replication value for future projects.

Scope of the claims

Dense text embeddings outperform classical and neural segmenters; candidate selection, not generation, is the dominant error source; prosodic and discourse signals over-segment relative to creator chapter granularity; CLIP is the one non-text signal that aligns with editorial chapter scale. That is the scope. What LECSEG-30 does *not* show: that the segmentations are pedagogically optimal, that students benefit measurably, or that findings transfer to annotation standards other than creator chapters.

6.4 Limitations

- **Corpus size and balance.** LECSEG-30 is a seed benchmark of 30 videos. The manifest covers five domains, but the distribution is not perfectly balanced.
- **Silver chapter labels.** Creator-provided YouTube chapters are useful but imperfect. They may reflect production choices rather than strictly pedagogical topic boundaries.
- **Conservative operating point.** The best P_k /WD method deliberately under-predicts boundaries to minimise false positives. This results in low diagnostic F1, which is expected and not a flaw: approximate segment structure and exact boundary recovery are different objectives, and the system is optimised for the former. A deployment that requires precise timestamp recovery would need a different operating point, likely at the cost of higher P_k .
- **Multimodal signal reliability.** Acoustic (pause/pitch), language-model perplexity, and BERTopic methods all produce worse P_k than BGE-divisive, reflecting a granularity mismatch. The exception is CLIP visual embeddings: CLIP alone reaches $P_k = 0.3958$, and CLIP plus text fusion reaches $P_k = 0.3740$, below the cross-model text method ($P_k = 0.3715$) but better than BGE-divisive ($P_k = 0.3884$). Visual scene changes in lecture slides carry chapter-level granularity and are a promising signal for future work.
- **LLM refinement.** The local LLM component is implemented, but the current verified metrics do not justify claiming large boundary-placement gains from LLM refinement.

6.5 Threats to validity

The following table summarises the principal threats and their estimated severity.

Table 6.1: Threat-to-validity overview for LECSEG.

Threat	Severity	Primary impact
Creator chapters as reference	High	Construct validity: benchmark measures creator metadata, not pedagogical truth
Small and imbalanced corpus	High	External validity: 30 videos limit generalisation claims
Single-domain annotator pool	Low–Medium	Annotation validity: both annotators share cultural and academic background; agreement may not generalise to other annotator groups
Gold boundary count in evaluation	Medium	Internal validity: deployment realism; real systems cannot know k in advance
Post-hoc method search	Medium	Internal validity: multiple testing risk mitigated by Holm correction
Domain imbalance (4 Math videos)	Medium	Selector stability on small domain splits
ASR quality not formally measured	Low–Medium	Upstream noise propagation from Whisper output

Construct validity. The benchmark evaluates a system’s ability to reproduce creator-provided navigation metadata, not pedagogically verified topic boundaries. As discussed in Section 4.2.4, creators may place chapter markers for editorial or discoverability reasons that do not coincide with genuine topic shifts. A system achieving low P_k is correctly predicting creator chapter structure; whether this predicts learning benefit requires an independent user study.

External validity. LECSEG-30 has 30 videos across five domains. Results should be treated as findings on this specific benchmark rather than as universal laws of lecture segmentation. The leave-domain-out selector degradation ($P_k = 0.4012$) is direct evidence that performance is sensitive to domain-matched training data; with more videos the picture may shift materially. A ten-video external spot-check across five domains (Section 5.12) found BGE-divisive consistent (external mean $P_k = 0.389$ vs 0.388 on LECSEG-30, $\Delta = 0.001$), while the cross-model method degraded somewhat ($P_k = 0.421$ vs 0.371, $\Delta = 0.050$). Confounds include yt-dlp auto-captions vs. Whisper transcripts and an external pool skewed toward 3Blue1Brown explainer videos.

Annotation validity. Subtopic labels were produced by human annotators and verified through independent double annotation. The inter-annotator agreement figure ($\kappa_{\text{subtopic}} = 0.4257$) represents genuine human agreement on a subjective

discourse-segmentation task, which is in the moderate range and appropriate for this type of annotation. The chapter-level $\kappa = 0.5351$ is inflated because both annotators derive chapter boundaries from the same YouTube creator metadata; the subtopic figure is the more meaningful validity measure.

Internal validity. Methods in the controlled evaluation receive the gold segment count. This isolates boundary placement, which is standard practice in topic segmentation, but it is not deployment-realistic. Boundary count error is reported as a diagnostic (Section 5.5), and automatic count prediction is listed as future work. The use of many candidate methods and operating points is mitigated by Holm correction over seven pre-specified comparisons, but the selector result should still be read as a low-resource operating point rather than a universal deployment guarantee.

Metric validity. P_k and WindowDiff correlate with navigation usefulness under the assumption that students find approximate-region boundary placement acceptable. This assumption is plausible but untested; a user study measuring actual navigation speed and retention would be needed to verify it empirically.

6.6 Closing the oracle gap

The oracle analysis is the clearest quantitative finding this thesis produces. With perfect candidate selection from the generated pool, $P_k = 0.0172$ is reachable at 96.8% recall. The gap between this oracle and the best deployable system ($\Delta P_k \approx 0.34$) indicates that, within the evaluated framework, the candidate pool already contains the information needed for near-perfect segmentation; the bottleneck appears to lie in correctly identifying which candidates to accept rather than in generating more candidates. This framing is the most actionable scientific output of the thesis: it shifts the research problem from *how to generate better candidates* to *how to rank candidates more reliably*.

Table 6.2 summarises the relevant operating points. TreeSeg (Gklezakos et al. 2024) reports $P_k = 0.367$ on TinyRec (21 self-recorded lectures), and LECSEG achieves $P_k = 0.3588$ with the balanced selector on LECSEG-30. These figures are indicative rather than directly comparable because the benchmarks, annotation sources, and video types all differ.

Three concrete reasons favour TreeSeg in any hypothetical direct comparison. First, it uses overlapping utterance-block embeddings, which provide richer contextual evidence per position than the single-sentence windows used in LECSEG. Second, TinyRec uses manually verified annotations rather than creator chapter metadata,

potentially making the reference labels more consistent. Third, LECSEG’s selector incurs leave-domain-out degradation that would not surface under same-domain evaluation. None of these diminish the LECSEG findings; they define the comparison boundary.

Table 6.2: Oracle gap analysis on LECSEG-30.

Operating point	$P_k \downarrow$	WD $\downarrow$
BGE-divisive baseline	0.3884	0.3956
Cross-model conservative (best global)	0.3713	0.3764
Balanced selector (best deployable)	0.3588	0.3739
TreeSeg paper (TinyRec, indicative only)	0.367	—
Per-video oracle (not deployable)	0.2980	0.3280
Candidate oracle $\tau = 2$ (upper bound)	0.0172	0.0198

The concrete research directions that could close this gap are:

1. **Supervised boundary-level ranker.** The video-level method selector shows that per-video method choice matters, but it does not rank individual boundary candidates. A ranker trained on candidate-level features (local cosine contrast, cross-model agreement, pause duration, slide-change signal, and OCR evidence) would directly address the selection bottleneck.
2. **Expanded annotation.** The current selector is trained leave-one-out on 30 videos. Expanding to 50–100 annotated lectures would provide $2\text{--}3\times$ more per-fold evidence and may substantially stabilise domain-specific selector behaviour.
3. **Shared-dataset evaluation.** Running TreeSeg’s original code on LECSEG-30 (or publishing LECSEG-30 as a community benchmark) would convert the current indicative comparison into a reproducible head-to-head result.
4. **CLIP-centred multimodal fusion.** CLIP is the only non-text signal that improves P_k /WD. A tighter integration of CLIP slide-scene evidence within the cross-model scoring framework, rather than as a secondary fusion component, could provide a more reliable visual prior for chapter-level transitions.

6.7 Lessons learned

Several lessons from this project are likely to carry over to related work.

Segmentation lessons.

1. **Dense text embeddings dominate at creator chapter granularity.** BGE and E5 cross-model scoring outperforms every multimodal combination tested. Semantic paragraph-level structure is the dominant signal for editorial-level chapter boundaries; fine-grained acoustic and lexical transitions are counterproductive.
2. **Auxiliary signals must be calibrated to the reference granularity, not just to transition detection.** Prosody and discourse markers detect real transitions, but those transitions are an order of magnitude finer than creator chapters. “Granularity mismatch” is a more precise diagnosis than “signal uninformative.”
3. **The generation/selection distinction is the key design choice.** Near-perfect boundaries are present in the candidate pool at 96.8% recall. Future systems should invest in selection and ranking, not in generating additional candidate positions.
4. **CLIP is the only tested non-text signal that aligns with editorial chapter granularity.** This makes it the most valuable non-text signal to integrate next, and the principled reason is the shared editorial origin of slide transitions and creator chapters, not a coincidence in the data.

Benchmark and annotation lessons.

5. **Small benchmarks enable failure analysis that large benchmarks cannot.** A 30-video, fully auditable corpus makes it possible to inspect every failure case, trace every metric value to its source video, and report negative results honestly. Large-scale leaderboards measure performance; small auditable benchmarks measure understanding.
6. **Creator chapters and pedagogical boundaries are different constructs.** Using creator chapters enables scale and reproducibility but couples the benchmark to editorial decisions. Any system trained or evaluated against creator chapters should be read as modelling navigation metadata rather than educational structure.
7. **Negative results have replication value.** Documenting that prosody, discourse markers, BERTopic, OCR, and local LLM refinement all fail to improve chapter-level P_k /WD prevents future projects from repeating these experiments without motivation. Most published theses omit failures; their inclusion here is a deliberate scientific choice.

6.8 Closing remarks

Lecture segmentation is far from solved, and this thesis does not pretend otherwise. What it contributes is a reproducible point of measurement and a clear diagnosis of where the remaining difficulty lies.

The two central findings — that the dominant error is in boundary selection not detection, and that many signals fail because they operate at the wrong discourse granularity — together define the research agenda for any successor system. A future design that addresses both, with a learned boundary ranker and signals explicitly calibrated to editorial chapter scale, has a clear, measurable target: close the $\Delta P_k \approx 0.34$ oracle gap.

There is also a broader point about benchmark scale worth making explicitly. Large benchmarks optimise for performance measurement: they reveal which system produces the best aggregate number on a large diverse test set. Small, auditable benchmarks optimise for *understanding*: they reveal why systems fail, which signals over- or under-segment, and where exactly the performance ceiling lies. LECSEG-30 was designed for the second purpose. The granularity mismatch finding and the oracle-gap diagnosis are both results that require the ability to inspect every prediction on every video — something a 200,000-video benchmark makes impractical. The value of LECSEG is not in competing with large-scale systems but in providing the kind of controlled, transparent analysis those systems cannot easily afford.

The LECSEG-30 corpus, annotation labels, evaluation code, and result files are open, so any future method can be evaluated against the same 30 videos under the same protocol. That reproducibility is what allows the findings here to be built upon rather than merely noted.

If there is one sentence in this thesis that carries the most weight, it is: *candidate generation is largely solved; candidate selection is the bottleneck*. That observation, supported by the oracle-gap analysis, tells the next researcher exactly where to invest effort — and it could only have been discovered by building the benchmark first.

6.9 Recommendations for Future Work

6.9.1 Robust candidate and method selection

The strongest immediate extension is a sequence-aware candidate selector. Current oracle and method-portfolio experiments show that plausible boundaries are often present in the candidate pool, but the system does not reliably pick the right subset

for each video. A better selector should combine local semantic contrast, cross-model agreement, pause/shot/OCR evidence, position priors, and global duration constraints while optimising directly against P_k , WindowDiff, and tolerance-F1.

6.9.2 Independent validation of chapter references

The most important dataset extension is not simply adding more videos; it is checking whether creator-provided YouTube chapters agree with independent human judgements. A focused validation study should select 8–10 lectures and ask two annotators to mark chapter-level boundaries without seeing the YouTube timestamps. The study should report human–human agreement, human–YouTube agreement, and system–human agreement under F1@2, F1@5, and time-tolerant metrics. This would quantify how reliable the current creator chapters actually are as reference boundaries.

6.9.3 Expanded annotator pool and domain diversity

The current inter-annotator study involves annotators from a shared academic background. Future work should recruit annotators from different disciplines and experience levels to test whether subtopic agreement generalises beyond the current pool. Annotating 8–10 additional videos with three or more independent annotators would give a Krippendorff’s α estimate alongside the existing Cohen’s κ , and would allow pairwise agreement patterns to be analysed rather than collapsed to a single mean figure.

6.9.4 External and Math-focused validation

The current selector fails most clearly on Mathematics, where the dataset has only four videos. A targeted extension should add 8–10 Math lectures with creator chapters and re-evaluate the final methods without retuning. If Math performance improves, the likely cause was training scarcity; if it stays weak, the problem probably runs deeper, involving mathematical notation, stable vocabulary, or gradual conceptual transitions. A separate 5–10 video external test set from new channels should also be used to measure generalisation beyond LECSEG-30.

6.9.5 Reliability-aware prosody and visual integration

Prosodic features (pause duration, pitch reset) and visual streams (shot boundaries, slide OCR) are available in the pipeline, but current ablations show that acoustic and linguistic sub-sentence signals can hurt P_k /WindowDiff when their granularity does not match chapter-level editorial decisions. The exception is CLIP visual

embeddings: CLIP alone achieves $P_k = 0.3958$ and CLIP plus text fusion reaches $P_k = 0.3740$, approaching the cross-model text method ($P_k = 0.3715$). Visual scene changes in lecture slides appear to carry chapter-level granularity. Future work should prioritise CLIP visual integration within the cross-model scoring framework, and apply quality-gating only to the noisier acoustic and OCR modalities.

6.9.6 Streaming and online inference

The current pipeline processes a complete video offline. A streaming variant would segment lecture content as it is being recorded, enabling real-time chaptering for live classes. This requires adapting the sliding-window fusion to operate on a causal context window and replacing the global threshold computation with an exponential moving average of past gap scores. The target latency budget for live chaptering is under 30 seconds of delay.

6.9.7 Cross-lingual extension

LECSEG currently targets English lectures. Extending to other languages requires: (1) replacing `all-mpnet-base-v2` with a multilingual text encoder such as `LaBSE` or `multilingual-e5-large`; and (2) using Whisper’s multilingual mode for transcription. Suitable evaluation data could come from non-English MOOCs (e.g., Coursera’s French and Spanish tracks) or from university recordings in South and South-East Asian languages.

6.9.8 End-to-end fine-tuning

The current architecture uses frozen pre-trained encoders. Fine-tuning the text encoder jointly with the boundary predictor on a larger annotated corpus (e.g., constructed by scaling the LECSEG-30 annotation procedure to 200+ videos) could yield further gains, at the cost of reduced generalisability. A natural training objective is a soft version of the tolerance- F_1 metric, differentiable via a sigmoid boundary-probability output.

6.9.9 Larger and domain-tuned local LLMs

Replacing Llama-3.1-8B with a 70B parameter model or a domain-specific fine-tune (e.g., fine-tuned on lecture transcripts) could improve title quality and boundary-shift accuracy, particularly for highly technical content. Quantised models (GGUF 4-bit) make 70B inference feasible on a single 24 GB GPU with acceptable latency (around 5 seconds per boundary refinement call).

6.9.10 Multi-speaker lectures and panels

Panel discussions and multi-instructor courses introduce speaker turns as an additional boundary signal. Speaker diarisation (e.g., using `pyannote.audio`) could provide a speaker-change feature that complements text embeddings, particularly for sections where the topic does not change but the speaker does.

6.9.11 User study on learning outcomes

All metrics in this thesis are information-retrieval proxies for segmentation quality. A controlled user study would directly measure whether learners locate target concepts faster, and whether their retention improves, when using LECSEG-generated chapters compared to flat video scrubbing or manually authored chapters. Such a study could also reveal which granularity (chapter vs. subtopic) is most useful for different learning tasks (review vs. first-watch).

6.9.12 Public benchmark and model release

A complete public release should include the manifest, video identifiers, creator chapter references, reviewed subtopic labels, preprocessing scripts, evaluation scripts, environment lock files, result-regeneration commands, and dataset/model cards. The current repository already contains the major components; the remaining work is release hygiene, privacy review, and a single documented reproduction command for the final thesis tables.

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Appendix A

Dataset Details

Appendix B

Dataset Details

B.1 Video list

Table B.1 lists the 30 videos in LECSEG-30. Full metadata, URLs, and timestamps are stored in `data/manifest.jsonl`, `data/gt/*.json`, and `data/gt_hier/*.json`.

Table B.1: LECSEG-30 video list, sorted by domain and video ID.

Video ID	Domain	Duration (min)	Chapters
9N1MxkbFhsc	Biology	42.8	23
AMl6E4cLrwE	Biology	45.6	13
KIVHqq38KJU	Biology	38.8	15
NNnIGh9g6fA	Biology	57.2	16
oOya3cFmAMc	Biology	51.7	15
uO5k9xcD1gU	Biology	47.3	12
D8RRq3TbtHU	CS	114.5	12
HtSuA80QTyo	CS	53.4	9
JP7ITIXGpHk	CS	105.6	26
Rl0ludWTLxs	CS	149.5	21
TjZBTDzGeGg	CS	47.3	11
jGwO_UgTS7I	CS	75.3	7
oqRU2So6Z2Y	CS	135.6	12
7K1sB05pE0A	Math	51.5	13
fsLh-NYhOoU	Math	60.4	13
j0wJBEZdwLs	Math	34.7	8
lUUte2o2Sn8	Math	65.2	21
8yT4RZy1t3s	Philosophy	55.2	18
GR63MMAi-fs	Philosophy	45.8	5
KNwMiydCYA4	Philosophy	63.4	25
MGyygiXMzRk	Philosophy	55.0	11
Qw4l1w0rkjs	Philosophy	55.1	34
S7TUe5w6RHo	Philosophy	63.5	29
Hy7ou5R_vjE	Physics	66.8	6
KOKnWaLiL8w	Physics	73.4	6
NK-BxowMIfg	Physics	66.0	5
Ns6GB4Dph9U	Physics	68.4	6
YdOXS_9_P4U	Physics	28.6	7
uK2eFv7ne_Q	Physics	73.8	6
zNVQfWC_evlg	Physics	60.3	14
Total	5 domains	1951.4	419

B.2 Domain distribution

Table B.2: LECSEG-30 domain distribution.

Domain	Videos	Duration (hours)	Chapters
Biology	6	4.72	94
CS	7	11.35	98
Math	4	3.53	55
Philosophy	6	5.63	122
Physics	7	7.29	50
Total	30	32.52	419

B.3 Subtopic annotation guidelines

The following guidelines were given to annotators for the subtopic annotation pass after chapter boundaries were fixed from YouTube metadata:

1. **Boundary definition.** A subtopic boundary occurs when the speaker transitions from one coherent pedagogical sub-idea to another within a chapter.
2. **Minimum duration.** Each subtopic segment should be at least 60 seconds unless the source chapter is shorter than that.
3. **Number per chapter.** Annotators were encouraged to use 2–5 subtopics per chapter when the chapter length supported it.
4. **Title format.** Titles should be short noun phrases, preferably using the speaker’s own terminology.
5. **Uncertainty.** When unsure, the annotator marked the best plausible transition. Both annotators worked independently; disagreements were preserved for IAA measurement rather than resolved by adjudication.

B.4 Inter-annotator agreement details

The IAA report covers all 30 videos and is stored in `data/gt_hier/iaa_report.json`. Subtopic boundaries were annotated independently by two human annotators; chapter boundaries are fixed from YouTube creator metadata (see Section 4.2.5 for full interpretation). Mean chapter-level $\kappa = 0.5351$ (inflated: both annotators observe the same creator timestamps) and mean subtopic-level $\kappa = 0.4257$ at tolerance 1 sentence. Boundary F1 is 1.0000 for chapters (expected, as noted above) and 0.7793 for subtopics.

Appendix C

Hyperparameters

Appendix D

Hyperparameters and System Configuration

All hyperparameters are stored in `configs/defaults.yaml` and can be overridden per experiment using Hydra. Every evaluation run records its exact configuration to `results/<run>/config.yaml`, ensuring full reproducibility.

D.1 Default configuration

D.2 Hardware and wall-clock runtimes

Experiments were run across the local Windows workstation and rented vast.ai GPU instances. Exact CPU/RAM values varied by run; GPU-backed runs used NVIDIA RTX-class hardware.

- CPU: local Windows workstation CPU for orchestration and light tests
- GPU: NVIDIA RTX 5090 for transcription; NVIDIA RTX 3090 for later embedding experiments
- RAM: instance-dependent; no run required model-parallel execution
- OS: Windows 11 locally; Ubuntu Linux on GPU instances

Approximate wall-clock times per pipeline stage for a 60-minute lecture:

The dominant cost is speech recognition; all other stages combined run in under two minutes per hour of video.

Table D.1: Key hyperparameters used in all reported experiments.

Component	Parameter	Value
ASR	Model	Whisper large-v3
	Beam size	1 (speed)
	Compute type	float16
	VAD filter	Enabled (500 ms min silence)
Text encoder	Backbone	BAAI/bge-large-en-v1.5 (primary)
	Embedding dimension	1024
	L2-normalised	Yes
Visual encoder	Backbone	CLIP ViT-B/32
	Embedding dimension	512
	Keyframe rate	1 fps
Two-stage predictor	Stage-1 threshold c_1	0.5σ below mean
	Stage-2 threshold c_2	1.2σ below mean
	Context window w	3 sentences
Reliability weighting	Fusion mode	Entropy-based
	Modalities	Text, visual, prosody
LLM refinement	Model	Llama-3.1-8B (Q4_K_M)
	Snap window	± 2 sentences
	Title max words	8
Evaluation	Bootstrap samples	1,000
	Confidence level	95 %
	Significance test	Wilcoxon + Holm correction
	Tolerance	± 2 sentences
Random seed	All stochastic ops	42

Table D.2: Wall-clock time per pipeline stage for a representative 60-minute lecture.

Stage	Time (seconds)
Whisper large-v3 transcription	≈ 200
Sentence splitting (spaCy)	< 1
Text embedding (BGE-large, CPU)	≈ 60
CLIP visual embedding (GPU, 1 fps)	≈ 60
RWF fusion + boundary prediction	< 1
LLM refinement + titling (8B Q4)	≈ 30
Total	≈ 350

Appendix E

Additional Results

Appendix F

Additional Results

F.1 Per-domain breakdown

Table F.1 reports the alignment-audited joint P_k /WindowDiff cross-model method, broken down by domain.

Table F.1: Per-domain evaluation of the current best official method.

Domain	Videos	$P_k \downarrow$	WD $\downarrow$	F1 $\uparrow$
Biology	6	0.3965	0.4009	0.0475
CS	7	0.3297	0.3338	0.0000
Mathematics	4	0.3793	0.3867	0.0000
Philosophy	6	0.3959	0.4073	0.0663
Physics	7	0.3665	0.3665	0.0000
All domains	30	0.3713	0.3764	0.0237

F.2 Additional benchmark rows

Table F.2: Selected additional benchmark and diagnostic rows.

Method	$P_k \downarrow$	WD $\downarrow$	Source
TextTiling	0.6053	0.8978	eval_bge.json
C99	0.4219	0.4494	eval_bge.json
CosineSeg	0.4902	0.5392	eval_bge.json
KMeansSeg	0.6172	0.9986	eval_bge.json
BertSeg	0.4891	0.5403	eval_bge.json
BGE + divisive	0.3884	0.3956	eval_bge.json
Cross-model conservative	0.3713	0.3764	eval_alignment_sweep.json
LOO ExtraTrees selector	0.3588	0.3739	method_selector_experiment_trainrank_balanced.json
Per-video method oracle	0.2980	0.3280	method_portfolio_analysis.json
BERT-wiki zero-shot	0.4932	0.5397	eval_bert_wiki.json
Boundary classifier LOOCV	0.5803	0.9933	eval_boundary_clf.json

F.3 Representative cases

The per-video results expose three useful cases for interpreting aggregate metrics:

- Best case by P_k : oqRU2So6Z2Y (CS), $P_k = 0.1457$.
- Typical mid-range cases are documented in the stored per-video evaluation rows.
- Hard case: S7TUe5w6RHo (Philosophy), $P_k = 0.4975$.

Together with the selector-choice audit in Table 5.8, these cases show that gains are concentrated in particular lectures rather than uniformly distributed across the benchmark.